

ULG Triad Spec v0.5

PeerCompute + Eshkol + Moonlab Integration

2026-06-05

Contents

1	Part 0 — README / Executive Overview	4
1.1	Abstract	4
1.2	Files	4
1.3	v0.5 change summary	4
1.4	System context	5
1.5	One-sentence architecture	5
1.6	Source notes	5
2	Part 1 — System Spec	7
2.1	Abstract	7
2.2	1. Purpose and scope	7
2.2.1	In scope	7
2.2.2	Out of scope for v0.5	7
2.3	2. System roles	8
2.3.1	2.1 PeerCompute	8
2.3.2	2.2 Eshkol	8
2.3.3	2.3 Moonlab	8
2.3.4	2.4 ULG carrier runtime	8
2.4	3. Core design rules	9
2.5	4. Runtime topology	9
2.6	5. Compute service manifest	10
2.7	6. Task capsule	10
2.8	7. Child-worker leases	11
2.9	8. GPU lease and shared GPU ABI	12
2.10	9. Artifacts	13
2.10.1	2.10.19.1 Quantum response artifact	13
2.10.2	2.10.29.2 Closure artifact	13
2.11	10. Provenance block	14
2.12	11. Validation report	14
2.13	12. Closure factory flow	15
2.14	13. Security and resource model	15
2.15	14. Conformance tests	16
2.16	Source notes	16
3	Part 2 — Architecture Plan	17
3.1	Abstract	17
3.2	1. Architectural thesis	17
3.3	2. System context	18
3.4	3. Logical layers	18
3.4.1	Layer A — PeerCompute orchestration	18
3.4.2	Layer B — Compute services	19
3.4.3	Layer C — Shared ABI and artifacts	19

3.4.4	Layer D — Hot execution	19
3.5	4. Worker-tree architecture	20
3.5.1	Rules	20
3.6	5. PeerCompute component additions	20
3.6.1	5.1 ComputeServiceRegistry	20
3.6.2	5.2 WorkerSupervisor	20
3.6.3	5.3 ChildWorkerLeaseManager	21
3.6.4	5.4 GpuBroker	21
3.6.5	5.5 ArtifactCache and ClosureRegistry	21
3.7	6. Eshkol service architecture	21
3.8	7. Moonlab service architecture	22
3.9	8. Shared GPU ABI	23
3.10	9. Data lifecycle	24
3.10.1	Hot layer	24
3.10.2	Warm layer	24
3.10.3	Cold layer	24
3.11	10. End-to-end execution flows	24
3.11.1	10.1 Closure derivation flow	25
3.11.2	10.2 ULG simulation flow	26
3.11.3	10.3 Task lifecycle	27
3.12	11. Validation and provenance	28
3.13	12. Visualization architecture	29
3.14	13. Architecture risks	29
3.15	Source notes	29
4	Part 3 — Implementation Roadmap	30
4.1	Abstract	30
4.2	1. Roadmap philosophy	30
4.3	2. Roadmap overview	31
4.4	3. Milestone 0.5 — specification package	31
4.5	4. Milestone 0.6 — shared ULG IR and GPU ABI	31
4.6	5. Milestone 0.7 — PeerCompute service orchestration	32
4.7	6. Milestone 0.8 — PeerCompute GPU broker and artifact cache	32
4.8	7. Milestone 0.9 — Eshkol closure service	32
4.9	8. Milestone 1.0 — Moonlab quantum service	33
4.10	9. Milestone 1.1 — Moonlab → Eshkol closure pipeline	33
4.11	10. Milestone 1.2 — EOS closure pipeline	33
4.12	11. Milestone 1.3 — distributed visualization and telemetry	34
4.13	12. Milestone 1.4 — hardening	34
4.14	13. First three demos	34
4.14.1	Demo A — Supervised service smoke test	34
4.14.2	Demo B — Quantum-derived potential	34
4.14.3	Demo C — Hydrogen/plasma closure	35
4.15	14. Definition of done for v1-compatible integration	35
4.16	Source notes	35
5	Part 4 — Library Extension Plan	36
5.1	Abstract	36
5.2	1. Extension strategy	36
5.3	2. PeerCompute extension plan	37
5.3.1	2.1 New packages/modules	37
5.3.2	2.2 ComputeServiceRegistry	38
5.3.3	2.3 Worker supervision	38
5.3.4	2.4 GPU broker	38
5.3.5	2.5 Artifact and closure cache	39
5.3.6	2.6 NetViz extensions	39
5.4	3. Eshkol extension plan	39
5.4.1	3.1 New capabilities	39

5.4.2	3.2 Example Eshkol ULG closure shape	39
5.4.3	3.3 Eshkol outputs	40
5.4.4	3.4 Eshkol service tasks	40
5.4.5	3.5 Eshkol acceptance tests	40
5.5	4. Moonlab extension plan	40
5.5.1	4.1 New capabilities	40
5.5.2	4.2 Moonlab service tasks	41
5.5.3	4.3 Moonlab artifact shape	41
5.5.4	4.4 Moonlab WebGPU priorities	41
5.5.5	4.5 Moonlab acceptance tests	42
5.6	5. Shared ULG GPU ABI package	42
5.6.1	Required descriptors	42
5.7	6. Cross-repository integration plan	42
5.7.1	Step 1 — publish ABI package	42
5.7.2	Step 2 — dummy service integration	42
5.7.3	Step 3 — real service bootstraps	42
5.7.4	Step 4 — real child-worker leases	42
5.7.5	Step 5 — real artifacts	42
5.7.6	Step 6 — ULG consumes closure	43
5.8	7. Backward compatibility	43
5.9	8. Repository ownership boundaries	43
5.10	9. What not to build	43
5.11	Source notes	43
6	Appendix A — Standalone Mermaid Diagram Sources	44
6.1	01_system_context.mmd	44
6.2	02_supervised_worker_tree.mmd	45
6.3	03_closure_factory_flow.mmd	46
6.4	04_shared_gpu_abi.mmd	47
6.5	05_data_lifecycle.mmd	48
6.6	06_task_state_machine.mmd	49
6.7	07_validation_provenance_loop.mmd	50
6.8	08_library_extension_map.mmd	51
6.9	09_roadmap.mmd	52
6.10	10_visualization_telemetry.mmd	52
6.11	11_security_resource_model.mmd	52
6.12	12_end_to_end_demo.mmd	53
6.13	Diagram index notes	53
6.14	01 System Context	54
6.15	02 Supervised Worker Tree	55
6.16	03 Closure Factory Flow	56
6.17	04 Shared Gpu Abi	57
6.18	05 Data Lifecycle	58
6.19	06 Task State Machine	59
6.20	07 Validation Provenance Loop	60
6.21	08 Library Extension Map	61
6.22	09 Roadmap	62
6.23	10 Visualization Telemetry	62
6.24	11 Security Resource Model	62
6.25	12 End To End Demo	63
7	Appendix B — JSON Schema Sketches	63
7.1	Schema index notes	64
7.2	closure_artifact.schema.json	64
7.3	compute_service_manifest.schema.json	65
7.4	quantum_response_artifact.schema.json	66
7.5	task_capsule.schema.json	67
7.6	validation_report.schema.json	68

1 Part 0 — README / Executive Overview

Version: **0.5**

Date: **2026-06-05**

1.1 Abstract

This v0.5 specification defines a combined browser-native scientific-computing stack built from **PeerCompute**, **Eshkol**, **Moonlab**, and a Universal Law Graph (**ULG**) simulation runtime. The goal is to let distributed browser peers run multiscale physics workloads where low-level quantum and mathematical tasks generate validated closures that are consumed by scalable WebGPU simulation kernels.

For readers unfamiliar with the three libraries: **PeerCompute** is treated as the browser-based P2P orchestration layer. It owns session topology, distributed task scheduling, shared hot/warm/cold state, compute-worker supervision, validation, artifact caching, and visualization. **Eshkol** is treated as a Scheme-like mathematical/law language for automatic differentiation, free-energy and response-function authoring, closure derivation, and WASM/WebGPU-compatible numerical compilation. **Moonlab** is treated as the quantum and many-body solver backend: it produces spectra, response samples, state-vector/tensor-network results, Hamiltonian evaluations, and CPU/WebGPU parity reports used by Eshkol and ULG. In this design, **PeerCompute is the authority**, while **Eshkol and Moonlab run as supervised WASM/WebGPU compute services** that may spawn child workers only through PeerCompute leases.

The plan assumes all three libraries will be extended. PeerCompute gains service registration, worker-tree supervision, GPU/resource leasing, closure/artifact registries, and NetViz extensions. Eshkol gains ULG law macros, closure manifests, derivative metadata, Moonlab task invocation, WGSL/table-generation targets, and WASM reference exports. Moonlab gains ULG quantum-task APIs, response-artifact schemas, WebGPU kernel expansion, and deterministic CPU/GPU parity reporting. A shared **ULG GPU ABI** connects all three so tensors, complex numbers, closure tables, worker messages, tolerance reports, and provenance blocks have one common representation.

1.2 Files

1. 01_SYSTEM_SPEC.md — normative runtime, service, artifact, and resource-control specification.
2. 02_ARCHITECTURE_PLAN.md — component architecture, execution flows, and data lifecycle.
3. 03_IMPLEMENTATION_ROADMAP.md — phased implementation plan with acceptance tests.
4. 04_LIBRARY_EXTENSION_PLAN.md — concrete extension plan for PeerCompute, Eshkol, and Moonlab.
5. COMBINED_SPEC.md — all documents merged into one Markdown file.
6. [diagrams/](#) — standalone Mermaid source files.
7. [schemas/](#) — initial JSON schema sketches.

1.3 v0.5 change summary

- Adds a formal abstract for readers unfamiliar with PeerCompute, Eshkol, Moonlab, and ULG.
- Promotes PeerCompute from generic compute host to the explicit orchestration authority.
- Defines Moonlab and Eshkol as supervised WASM/WebGPU compute services that may spawn child workers only through PeerCompute leases.
- Adds a shared ULG GPU ABI and ULG IR boundary so all three libraries exchange tensors, complex arrays, closure tables, kernel modules, tolerance reports, and provenance blocks consistently.
- Adds schema sketches for service manifests, task capsules, closure artifacts, GPU leases, and validation reports.
- Expands the plan for extending all three libraries.

1.4 System context

PeerCompute + Eshkol + Moonlab: ULG Orchestration Context

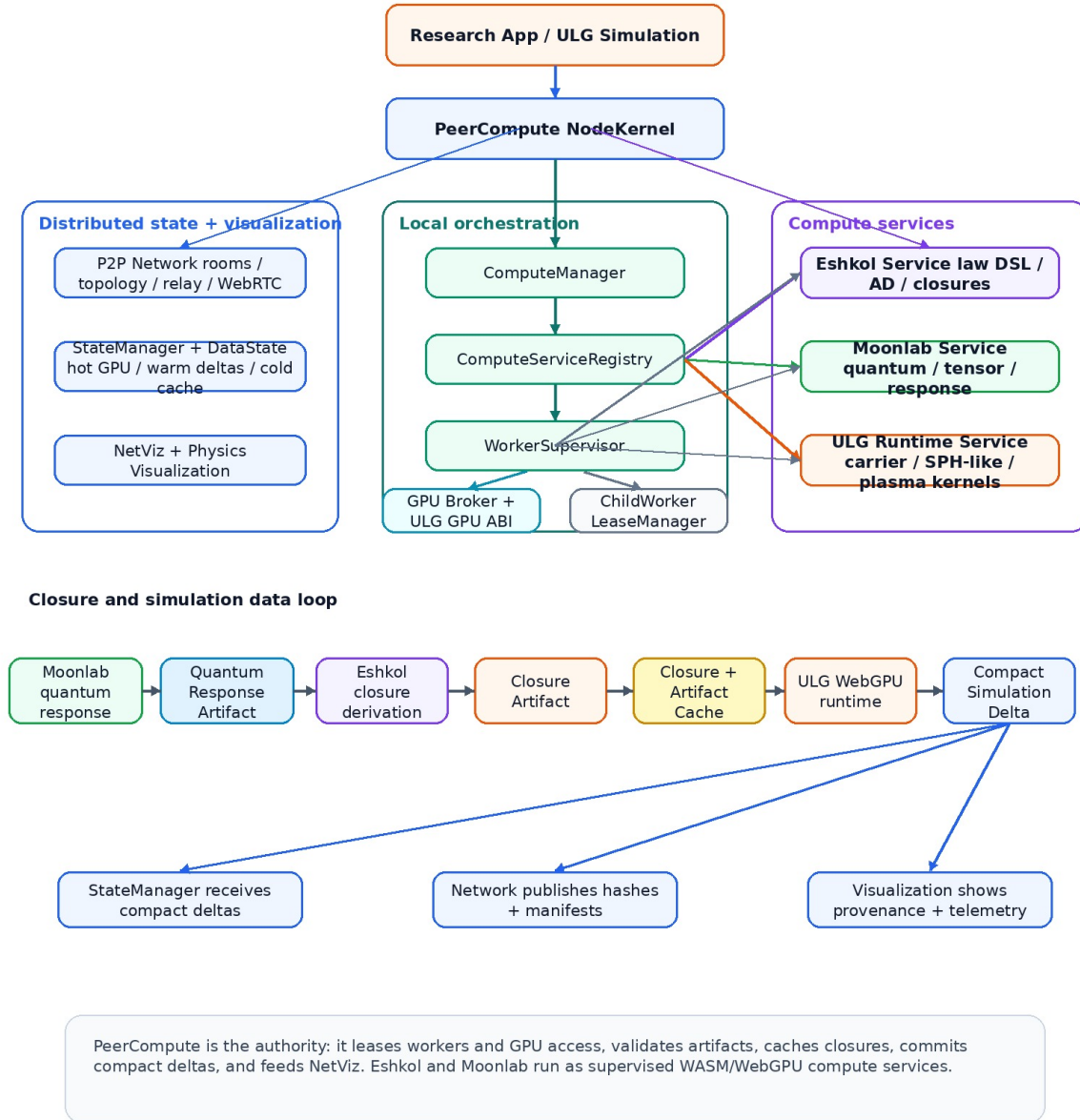


Figure 1: System context and orchestration loop

1.5 One-sentence architecture

PeerCompute supervises distributed worker trees and visualization; Eshkol derives and compiles physics closures; Moonlab supplies quantum/many-body response artifacts; ULG/WebGPU workers consume validated closures to run scalable carrier simulations.

1.6 Source notes

This specification is based on the public project documentation checked on 2026-06-05:

- [PeerCompute repository and README](#)

- [Eshkol repository and README](#)
- [Moonlab repository and README](#)
- [Moonlab architecture notes](#)
- [Moonlab WebGPU plan](#)
- [Web Workers documentation](#)
- [Worker WebGPU entry point](#)
- [SharedArrayBuffer isolation requirements](#)
- [WGSL specification](#)

2 Part 1 — System Spec

Version: **0.5**

Date: **2026-06-05**

Status: **draft specification**

2.1 Abstract

This v0.5 specification defines a combined browser-native scientific-computing stack built from **PeerCompute**, **Eshkol**, **Moonlab**, and a Universal Law Graph (**ULG**) simulation runtime. The goal is to let distributed browser peers run multiscale physics workloads where low-level quantum and mathematical tasks generate validated closures that are consumed by scalable WebGPU simulation kernels.

For readers unfamiliar with the three libraries: **PeerCompute** is treated as the browser-based P2P orchestration layer. It owns session topology, distributed task scheduling, shared hot/warm/cold state, compute-worker supervision, validation, artifact caching, and visualization. **Eshkol** is treated as a Scheme-like mathematical/law language for automatic differentiation, free-energy and response-function authoring, closure derivation, and WASM/WebGPU-compatible numerical compilation. **Moonlab** is treated as the quantum and many-body solver backend: it produces spectra, response samples, state-vector/tensor-network results, Hamiltonian evaluations, and CPU/WebGPU parity reports used by Eshkol and ULG. In this design, **PeerCompute is the authority**, while **Eshkol and Moonlab run as supervised WASM/WebGPU compute services** that may spawn child workers only through PeerCompute leases.

The plan assumes all three libraries will be extended. PeerCompute gains service registration, worker-tree supervision, GPU/resource leasing, closure/artifact registries, and NetViz extensions. Eshkol gains ULG law macros, closure manifests, derivative metadata, Moonlab task invocation, WGSL/table-generation targets, and WASM reference exports. Moonlab gains ULG quantum-task APIs, response-artifact schemas, WebGPU kernel expansion, and deterministic CPU/GPU parity reporting. A shared **ULG GPU ABI** connects all three so tensors, complex numbers, closure tables, worker messages, tolerance reports, and provenance blocks have one common representation.

2.2 1. Purpose and scope

This document specifies the runtime contracts for integrating PeerCompute, Eshkol, Moonlab, and ULG into a distributed browser-native scientific-computing system.

The design assumes all three upstream libraries will be extended. It does not assume Eshkol or Moonlab replace PeerCompute. PeerCompute remains the scheduling, network, state, validation, visualization, and local resource authority. Eshkol and Moonlab become compute services managed by PeerCompute.

2.2.1 In scope

- PeerCompute compute-service registration.
- Long-lived WASM/WebGPU service workers.
- Supervised child workers for Moonlab and Eshkol.
- GPU/resource leases.
- ULG GPU ABI and closure artifact contracts.
- Quantum response artifacts from Moonlab.
- Closure derivation artifacts from Eshkol.
- Validation, provenance, caching, visualization, and distributed publication.

2.2.2 Out of scope for v0.5

- A complete physics implementation.
- A complete DFT/QED/nuclear solver.
- Browser security bypasses or unsupervised remote code execution.

- Full-state replication of large tensors or simulation domains.
- Native-cluster orchestration outside PeerCompute’s browser-first model.

2.3 2. System roles

2.3.1 2.1 PeerCompute

PeerCompute is the orchestration authority. It hosts:

- NodeKernel session authority;
- P2P room/topology/network scheduling;
- shared state via hot/warm/cold DataState layers;
- compute service registry;
- worker supervisor and child-worker lease manager;
- GPU broker;
- artifact and closure cache;
- validation manager;
- task lineage tracker;
- NetViz and physics telemetry.

PeerCompute should evolve from a flat task runner into a **supervised distributed worker-tree runtime**.

2.3.2 2.2 Eshkol

Eshkol is the law and closure compiler. It hosts:

- ULG law macros;
- free-energy, response-function, and EOS definitions;
- symbolic/forward/reverse AD;
- derivative metadata export;
- closure-table generation;
- WASM reference functions;
- optional WGSL/table-interpolation kernel generation;
- Moonlab task invocation when closure derivation requires quantum response data.

Eshkol does not own P2P networking, visualization, distributed authority, or unsupervised GPU scheduling.

2.3.3 2.3 Moonlab

Moonlab is the quantum and many-body backend. It hosts:

- state-vector tasks;
- tensor-network tasks;
- Hamiltonian matvec and Lanczos/Krylov primitives;
- response and transition-sample tasks;
- quantum chemistry / VQE-style reference jobs where available;
- CPU/WebGPU parity checks;
- ULG-compatible quantum response artifacts.

Moonlab does not own the distributed simulation state, closure cache publication, or worker-tree authority.

2.3.4 2.4 ULG carrier runtime

The ULG runtime consumes validated closures and runs scalable simulation kernels:

- carrier integration;
- spatial hash / neighborhood construction;
- edge-message evaluation;
- SPH-like smoothing observers;

- plasma/MHD/radiation kernels;
- closure interpolation;
- invariant/reduction passes;
- compact delta emission.

The ULG runtime may be implemented as a first-party PeerCompute service or as a separately registered service.

2.4 3. Core design rules

1. **PeerCompute owns orchestration.** Eshkol and Moonlab are service workers, not network authorities.
2. **Nested workers are supervised.** Eshkol/Moonlab may spawn child workers only through PeerCompute-granted leases.
3. **GPU use is brokered.** Services request WebGPU leases from PeerCompute instead of independently competing with rendering and simulation kernels.
4. **Every result is provenance-bearing.** Artifacts must identify method, input hash, source service, assumptions, validity, uncertainty, and validation status.
5. **CPU/WASM remains the reference path.** WebGPU acceleration is accepted only with tolerance policies and parity testing.
6. **Shared ABI before performance work.** The ULG GPU ABI defines dtype layout, tensor layout, complex representation, closure table format, kernel descriptors, and tolerance reports.
7. **No opaque compute islands.** Worker trees, child tasks, GPU leases, memory pressure, validation status, and closure cache state are visible to PeerCompute and NetViz.
8. **No full-state replication by default.** Distributed tasks exchange hashes, manifests, compact deltas, artifacts, closure handles, and boundary data.

2.5 4. Runtime topology

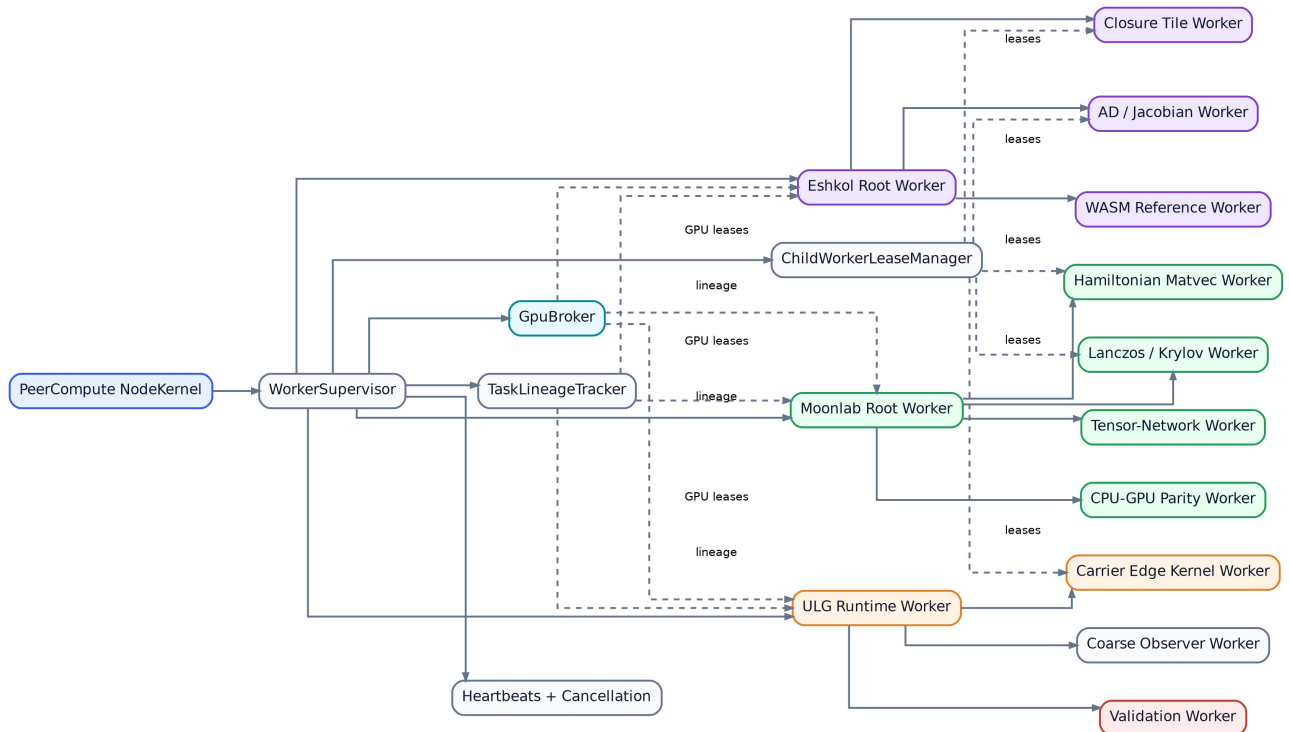


Figure 2: Supervised worker tree

2.6 5. Compute service manifest

Every service registers a manifest with PeerCompute.

```
type ComputeServiceManifest = {
  serviceId: "eshkol" | "moonlab" | "ulg-runtime" | string
  version: string
  runtime: "wasm" | "webgpu" | "wasm-webgpu" | "js" | "native-bridge"

  entry: {
    workerModule: string
    wasmModule?: string
    initExport?: string
  }

  childWorkers: {
    allowed: boolean
    maxChildren: number
    allowedModules: string[]
    sameOriginOnly: true
  }

  resources: {
    maxWasmMemoryBytes: number
    maxGpuMemoryBytes?: number
    maxCpuWorkers?: number
    maxTaskMs?: number
  }

  capabilities: string[]
  abi: {
    ulgIrVersion: string
    gpuAbiVersion: string
    supportedDTypes: string[]
    supportedLayouts: string[]
  }

  validation: {
    requiresCpuReference: boolean
    toleranceProfile: string
    parityModes: string[]
  }
}
```

2.7 6. Task capsule

All work is packaged as task capsules.

```
type ULGTaskCapsule = {
  taskId: string
  rootTaskId: string
  serviceId: string
  taskKind:
    | "closure.derive"
    | "closure.table.generate"
    | "quantum.response"
    | "quantum.spectrum"
    | "quantum.tensor_network"
    | "simulation.step"
```

```

    | "validation.reference"
    | "artifact.cache.publish"

inputs: ArtifactRef[]
outputs: ArtifactDecl[]
unitsHash: string
inputHash: string
methodHash: string
deterministicSeed?: string

resources: {
  childWorkers?: number
  gpu?: "none" | "optional" | "required"
  wasmMemoryBytes?: number
  gpuMemoryBytes?: number
  priority: "render" | "interactive" | "simulation" | "background" | "validation"
}

validation: {
  mode: "none" | "self" | "cpu-reference" | "quorum" | "spot-check"
  toleranceProfile: string
}

provenance: ProvenanceBlock
}

```

2.8 7. Child-worker leases

A root service worker requests child-worker leases from PeerCompute.

```

type ChildWorkerLease = {
  leaseId: string
  parentWorkerId: string
  rootTaskId: string
  module: string
  count: number
  expiresAt?: number
  resources: {
    wasmMemoryBytes: number
    cpuWeight: number
    gpu: boolean
    expectedDurationMs?: number
  }
  cancellationToken: string
}

```

A service must not call new Worker() for arbitrary modules. It must use leased, allowlisted, same-origin modules that PeerCompute can observe and cancel.

2.9 8. GPU lease and shared GPU ABI

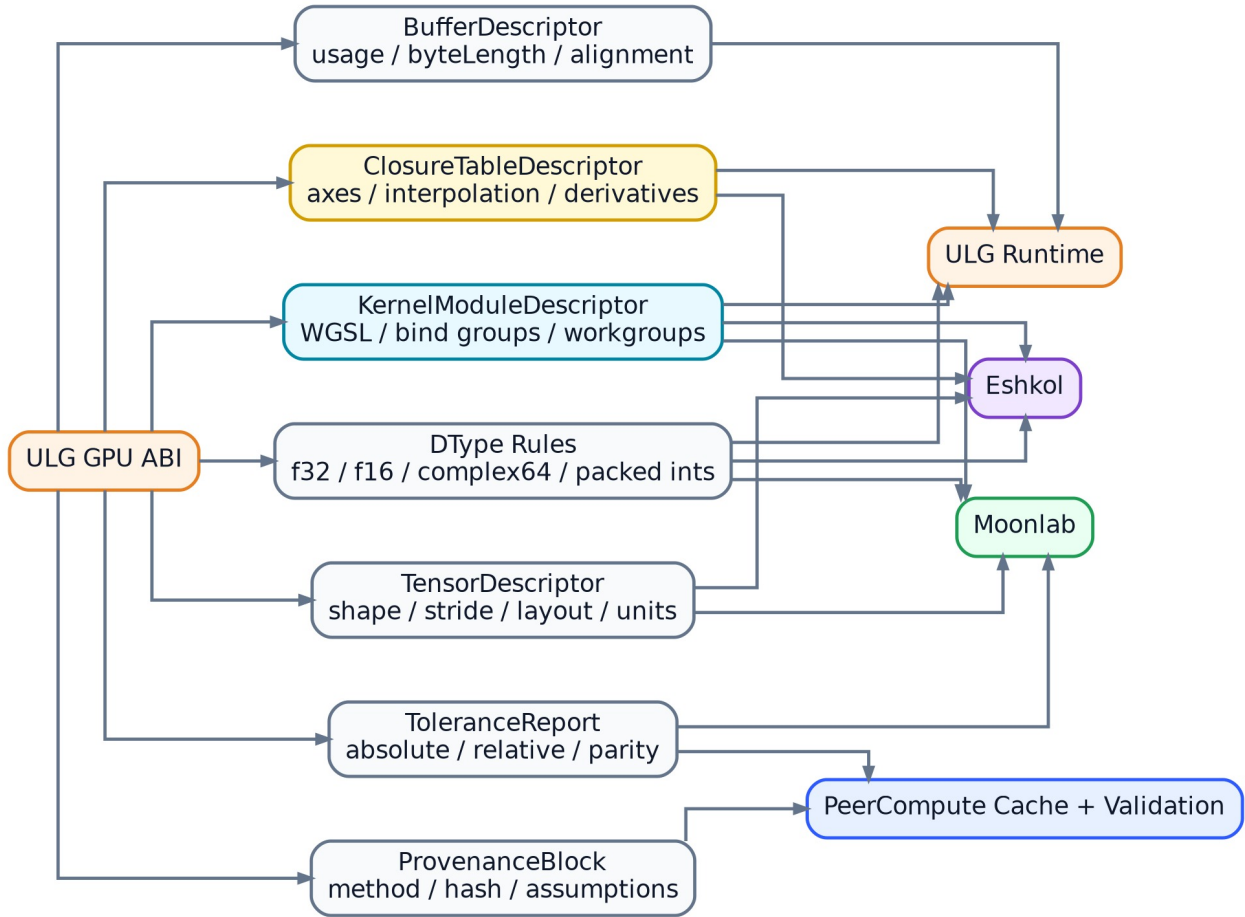


Figure 3: Shared ULG GPU ABI

```

type GpuLease = {
  leaseId: string
  serviceId: string
  rootTaskId: string
  priority: "render" | "interactive" | "simulation" | "background" | "validation"
  adapterFingerprint: string
  limitsHash: string
  maxBufferBytes: number
  maxEstimatedMemoryBytes: number
  expiresAt?: number
  cancellationToken: string
}

```

The ULG GPU ABI defines:

- scalar dtypes: f32, optional f16, packed integer types;
- complex layout: complex64 as either interleaved [re, im] or SoA with descriptor flag;
- tensor descriptors: shape, stride, dtype, layout, units hash;
- closure table descriptors: axis ranges, interpolation mode, derivative availability;
- kernel descriptors: WGSL module hash, bind-group layout, workgroup size, resource class;
- tolerance reports: absolute, relative, stochastic, CPU/GPU parity, drift.

2.10 9. Artifacts

2.10.1 9.1 Quantum response artifact

```

type QuantumResponseArtifact = {
  artifactId: string
  sourceService: "moonlab"
  taskKind: "quantum.response" | "quantum.spectrum" | "quantum.tensor_network"
  inputHash: string
  method: string
  basis?: string
  representation: "state_vector" | "tensor_network" | "density_matrix" | "operator" |
  ↪ "samples"
  outputs: {
    energyLevels?: ArtifactRef
    groundStateEnergy?: number
    transitionMatrixElements?: ArtifactRef
    densityOfStates?: ArtifactRef
    forceSamples?: ArtifactRef
    correlationFunctions?: ArtifactRef
  }
  uncertainty: UncertaintyReport
  validation: ValidationReport
  provenance: ProvenanceBlock
}

```

2.10.2 9.2 Closure artifact

```

type ClosureArtifact = {
  closureId: string
  sourceService: "eshkol" | "imported" | "manual"
  sourceLanguage?: "eshkol"
  closureKind:
    | "eos"
    | "opacity"
    | "interatomic_potential"
    | "transport"
    | "optical_response"
    | "material_tensor"
    | "reaction_rate"

  inputs: PortDecl[]
  outputs: PortDecl[]
  derivatives?: DerivativeDecl[]

  execution: {
    mode: "wasm-reference" | "wgsi-kernel" | "table-interpolation" | "hybrid"
    wasmRef?: ArtifactRef
    wgsiModule?: ArtifactRef
    table?: ArtifactRef
  }

  validity: ValidityEnvelope
  uncertainty: UncertaintyReport
  validation: ValidationReport
  provenance: ProvenanceBlock
}

```

2.11 10. Provenance block

```
type ProvenanceBlock = {  
  createdAt: string  
  createdByPeer: string  
  sourceRepo?: string  
  sourceCommit?: string  
  sourceService: string  
  sourceVersion: string  
  method: string  
  assumptions: string[]  
  inputHash: string  
  methodHash: string  
  artifactHash: string  
  parentArtifacts?: string[]  
}
```

2.12 11. Validation report

```
type ValidationReport = {  
  status: "unchecked" | "pass" | "warn" | "fail" | "quarantined"  
  validatorPeer?: string  
  validationMode: "self" | "cpu-reference" | "quorum" | "spot-check" | "invariant-audit"  
  toleranceProfile: string  
  maxAbsError?: number  
  maxRelError?: number  
  invariantDrift?: Record<string, number>  
  notes?: string[]  
}
```

2.13 12. Closure factory flow

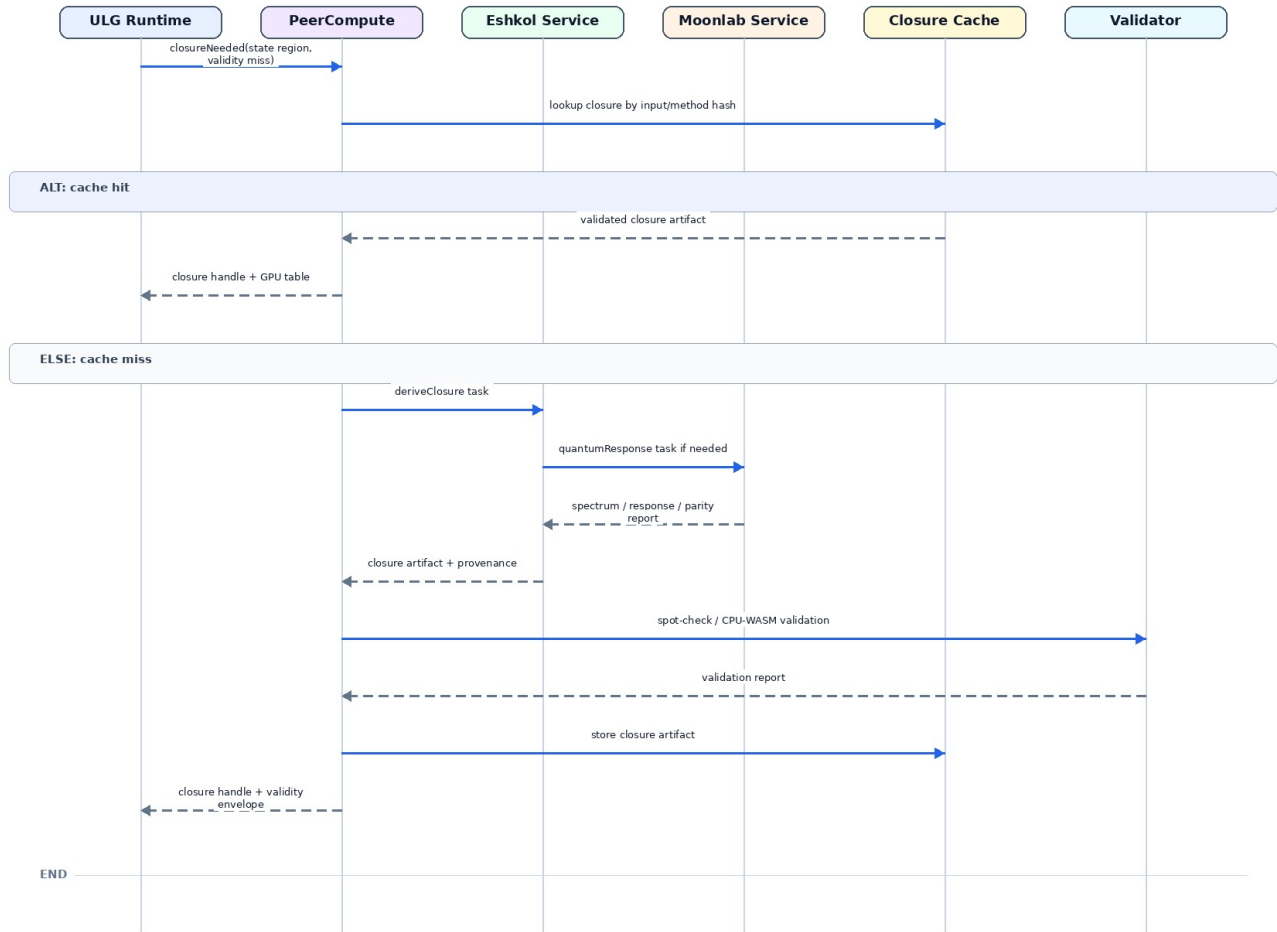


Figure 4: Closure derivation sequence

2.14 13. Security and resource model

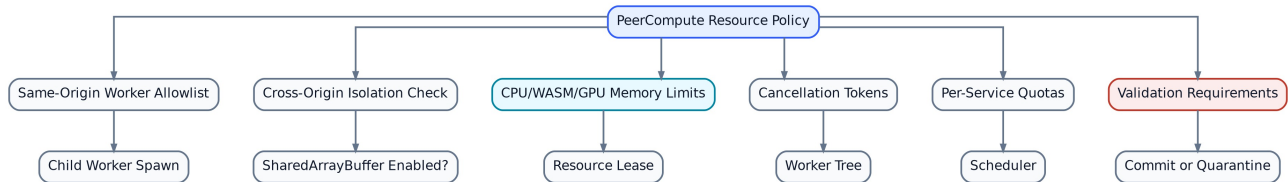


Figure 5: Resource and security policy

Requirements:

- Child worker modules must be same-origin and allowlisted.
- SharedArrayBuffer may be used only when cross-origin isolation is available.
- Large data should move through transferables, shared memory, IndexedDB artifact refs, or GPU buffers; avoid large structured clones.
- PeerCompute must be able to cancel a whole worker tree.

- WebGPU device-lost events must cause task retry, fallback, or quarantine.
- No service may publish distributed state directly. State commits go through PeerCompute.

2.15 14. Conformance tests

A v0.5-conforming prototype should pass:

1. Register Eshkol, Moonlab, and ULG services with manifests.
2. Spawn root service workers under PeerCompute.
3. Grant and revoke child-worker leases.
4. Grant and revoke GPU leases.
5. Run a Moonlab CPU/WebGPU parity task and emit a QuantumResponseArtifact.
6. Run an Eshkol closure derivation task and emit a ClosureArtifact.
7. Cache an artifact by input/method/artifact hash.
8. Use a cached closure in a ULG WebGPU kernel.
9. Show worker tree, GPU lease, cache, and validation telemetry in NetViz.
10. Cancel a root task and confirm all children stop.

2.16 Source notes

This specification is based on the public project documentation checked on 2026-06-05:

- [PeerCompute repository and README](#)
- [Eshkol repository and README](#)
- [Moonlab repository and README](#)
- [Moonlab architecture notes](#)
- [Moonlab WebGPU plan](#)
- [Web Workers documentation](#)
- [Worker WebGPU entry point](#)
- [SharedArrayBuffer isolation requirements](#)
- [WGSL specification](#)

3 Part 2 — Architecture Plan

Version: **0.5**

Date: **2026-06-05**

3.1 Abstract

This v0.5 specification defines a combined browser-native scientific-computing stack built from **PeerCompute**, **Eshkol**, **Moonlab**, and a Universal Law Graph (**ULG**) simulation runtime. The goal is to let distributed browser peers run multiscale physics workloads where low-level quantum and mathematical tasks generate validated closures that are consumed by scalable WebGPU simulation kernels.

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The plan assumes all three libraries will be extended. PeerCompute gains service registration, worker-tree supervision, GPU/resource leasing, closure/artifact registries, and NetViz extensions. Eshkol gains ULG law macros, closure manifests, derivative metadata, Moonlab task invocation, WGSL/table-generation targets, and WASM reference exports. Moonlab gains ULG quantum-task APIs, response-artifact schemas, WebGPU kernel expansion, and deterministic CPU/GPU parity reporting. A shared **ULG GPU ABI** connects all three so tensors, complex numbers, closure tables, worker messages, tolerance reports, and provenance blocks have one common representation.

3.2 1. Architectural thesis

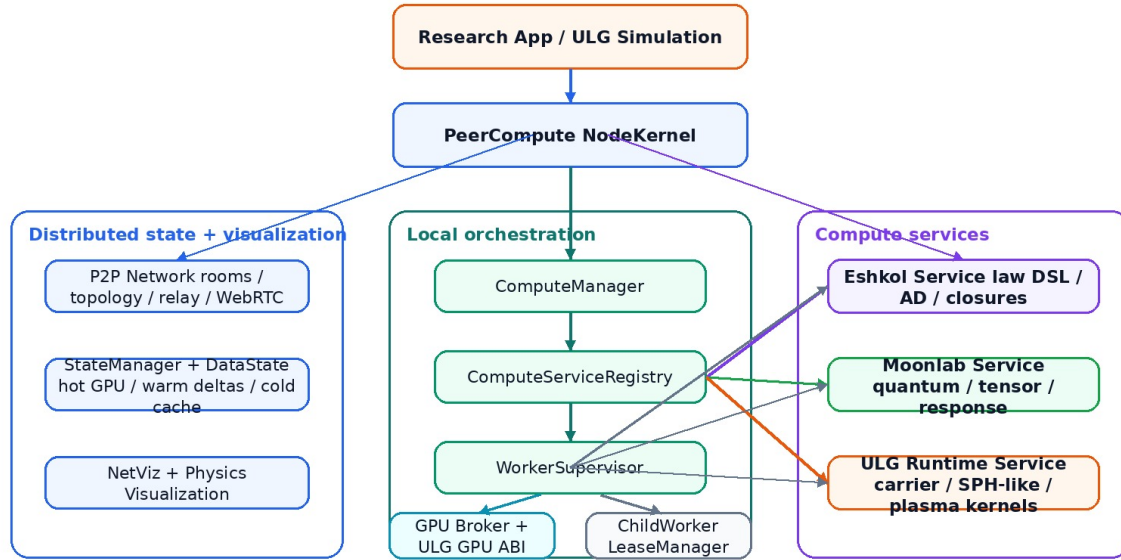
The combined system should be built as a **service-oriented browser scientific runtime**:

```
PeerCompute = authority, network, state, visualization, supervision
Eshkol      = mathematical law compiler and closure factory
Moonlab     = quantum/many-body response engine
ULG Runtime = scalable carrier simulation consuming validated closures
WebGPU/WGSL = hot dense numerical kernels
WASM/CPU    = reference and fallback execution
```

The key architecture choice is that Moonlab and Eshkol run inside PeerCompute's supervised compute tree. They may spawn child workers, but only through PeerCompute leases. This avoids opaque compute islands and lets the same orchestration layer handle distribution, visualization, validation, cancellation, cache publication, and resource pressure.

3.3 2. System context

PeerCompute + Eshkol + Moonlab: ULG Orchestration Context



Closure and simulation data loop

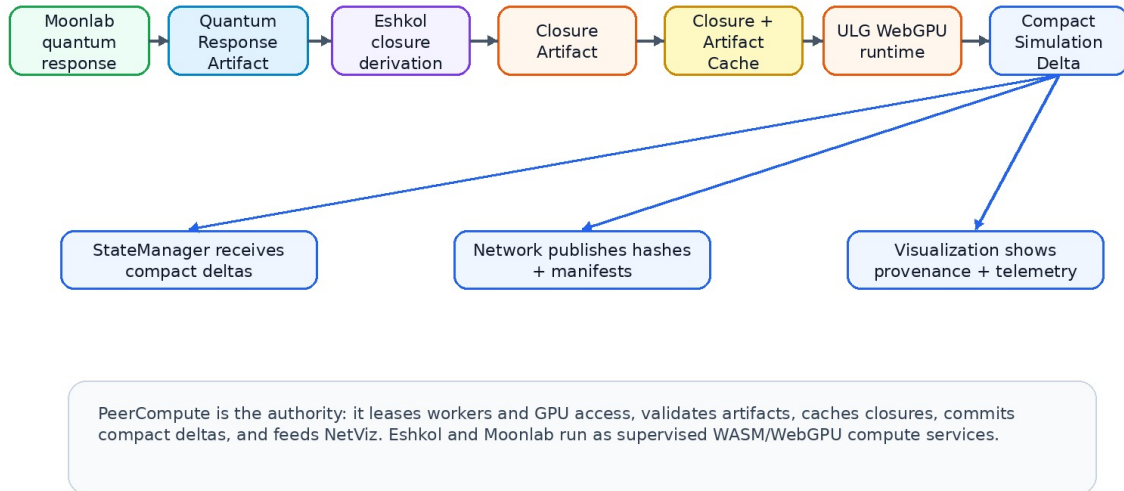


Figure 6: System context and orchestration loop

3.4 3. Logical layers

3.4.1 Layer A — PeerCompute orchestration

Responsibilities:

- peer discovery, rooms, relay/direct topology, and network state;
- distributed task scheduling;
- service registration and capability discovery;
- worker supervision and cancellation;
- GPU/resource leasing;

- hot/warm/cold data lifecycle;
- artifact cache and closure registry;
- validation/quorum/spot-checks;
- NetViz and physics telemetry.

3.4.2 Layer B — Compute services

Services registered under PeerCompute:

- eshkol: law DSL, AD, closure derivation, table generation, WASM/WGSL outputs.
- moonlab: state-vector/tensor-network/quantum response tasks.
- ulg-runtime: carrier, SPH-like, MHD, radiation, observer, and validation kernels.

3.4.3 Layer C — Shared ABI and artifacts

All services communicate through:

- ULG IR manifests;
- ULG GPU ABI;
- typed artifacts;
- validation reports;
- provenance blocks;
- closure cache records;
- task lineage metadata.

3.4.4 Layer D — Hot execution

Hot execution lives in WebGPU workers and WASM:

- WGSL compute kernels;
- WASM reference functions;
- child workers for parallel CPU/WASM tasks;
- optional shared-memory paths when available.

3.5 4. Worker-tree architecture

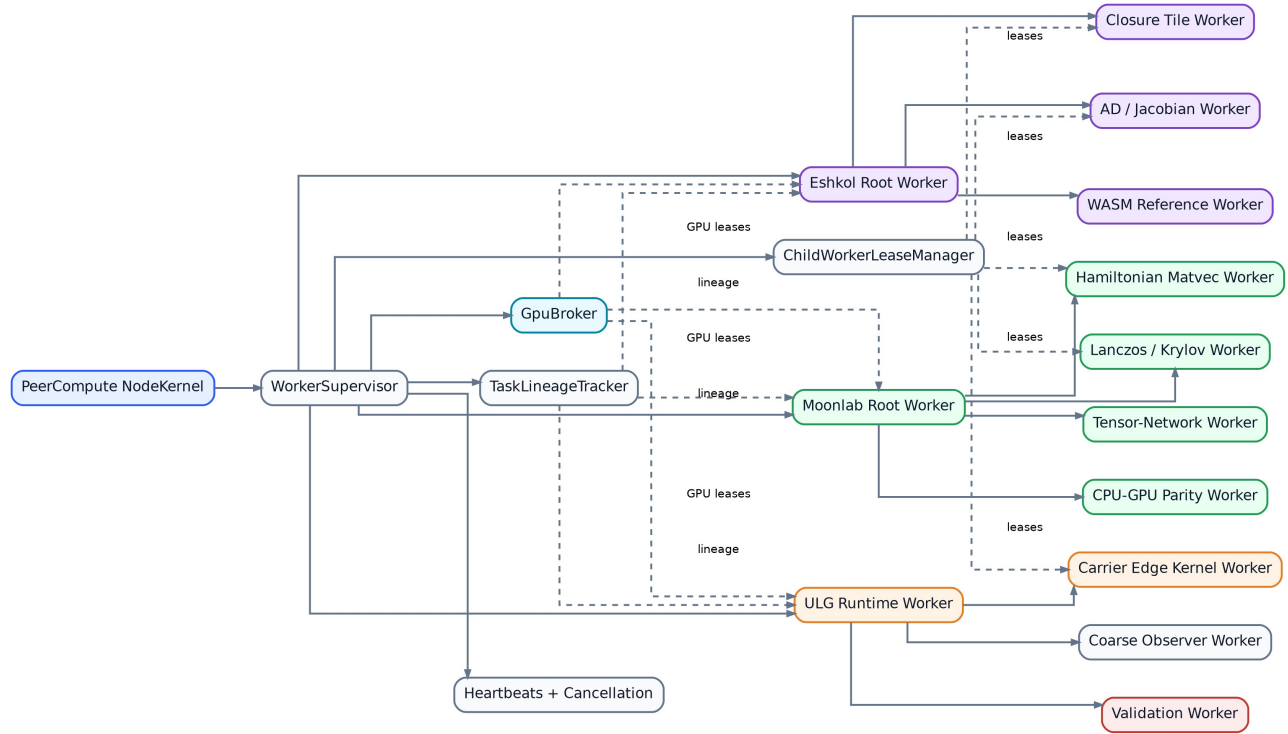


Figure 7: Supervised worker tree

3.5.1 Rules

1. Root workers are spawned by PeerCompute.
2. Child workers are spawned only with PeerCompute leases.
3. Each worker reports heartbeat, task status, memory estimates, GPU lease usage, and child status.
4. Parent service workers aggregate child results before returning to PeerCompute.
5. PeerCompute can cancel a root task and its entire subtree.

3.6 5. PeerCompute component additions

3.6.1 5.1 ComputeServiceRegistry

Loads service manifests and exposes capabilities.

```

interface ComputeServiceRegistry {
  register(manifest: ComputeServiceManifest): Promise<ServiceHandle>
  listCapabilities(): ServiceCapabilityIndex
  resolve(taskKind: string): ServiceHandle[]
}

```

3.6.2 5.2 WorkerSupervisor

Manages root workers, child workers, lifetimes, cancellation, health, and telemetry.

```

interface WorkerSupervisor {
  spawnService(serviceId: string): Promise<RootWorkerHandle>
  submitTask(task: ULGTaskCapsule): Promise<ComputeResult>
  cancelTree(rootTaskId: string): Promise<void>
}

```

```
getTreeTelemetry(rootTaskId: string): WorkerTreeTelemetry
}
```

3.6.3 5.3 ChildWorkerLeaseManager

Prevents runaway nested workers.

```
interface ChildWorkerLeaseManager {
  request(parentWorkerId: string, spec: ChildWorkerLeaseRequest):
    ↪ Promise<ChildWorkerLease>
  release(leaseId: string): Promise<void>
  revokeByRootTask(rootTaskId: string): Promise<void>
}
```

3.6.4 5.4 GpuBroker

Brokers WebGPU access across rendering, ULG simulation, Moonlab, and Eshkol.

```
interface GpuBroker {
  probe(): Promise<GpuCapabilities>
  requestLease(spec: GpuLeaseRequest): Promise<GpuLease>
  releaseLease(leaseId: string): Promise<void>
  reportPressure(): GpuPressureReport
}
```

Priority order should be:

```
render / UI > interactive simulation > simulation step > closure generation > quantum
↪ reference > background validation
```

3.6.5 5.5 ArtifactCache and ClosureRegistry

Artifacts are content-addressed by input hash, method hash, and artifact hash.

```
interface ArtifactCache {
  put(artifact: ArtifactRecord): Promise<ArtifactRef>
  get(ref: ArtifactRef): Promise<ArtifactRecord | undefined>
  announce(ref: ArtifactRef): Promise<void>
}

interface ClosureRegistry {
  resolve(query: ClosureQuery): Promise<ClosureArtifact | undefined>
  store(closure: ClosureArtifact): Promise<ArtifactRef>
  invalidate(reason: InvalidationEvent): Promise<void>
}
```

3.7 6. Eshkol service architecture

```
Eshkol Root Worker
├─ parser/macro expansion
├─ type/unit/dimensional metadata
├─ AD engine
├─ closure table generator
├─ WASM reference export
├─ WGSL/table strategy emitter
└─ Moonlab task client
```

└ child workers for tile generation and validation

Eshkol should expose service task kinds:

```
eshkol.law.compile
eshkol.closure.derive
eshkol.closure.table.generate
eshkol.derivative.export
eshkol.wasm.reference.build
eshkol.wgsl.emit
eshkol.validation.reference
```

3.8 7. Moonlab service architecture

Moonlab Root Worker

```
├ WASM module loader
├ WebGPU backend probe
├ state-vector task adapter
├ tensor-network task adapter
├ Hamiltonian/matvec task adapter
├ Lanczos/Krylov task adapter
├ response artifact exporter
└ CPU/GPU parity runner
```

Moonlab should expose service task kinds:

```
moonlab.quantum.response
moonlab.quantum.spectrum
moonlab.quantum.ground_state
moonlab.quantum.transition_matrix
moonlab.quantum.correlation
moonlab.tensor_network.contract
moonlab.tensor_network.expectation
moonlab.parity.cpu_webgpu
```

3.9 8. Shared GPU ABI

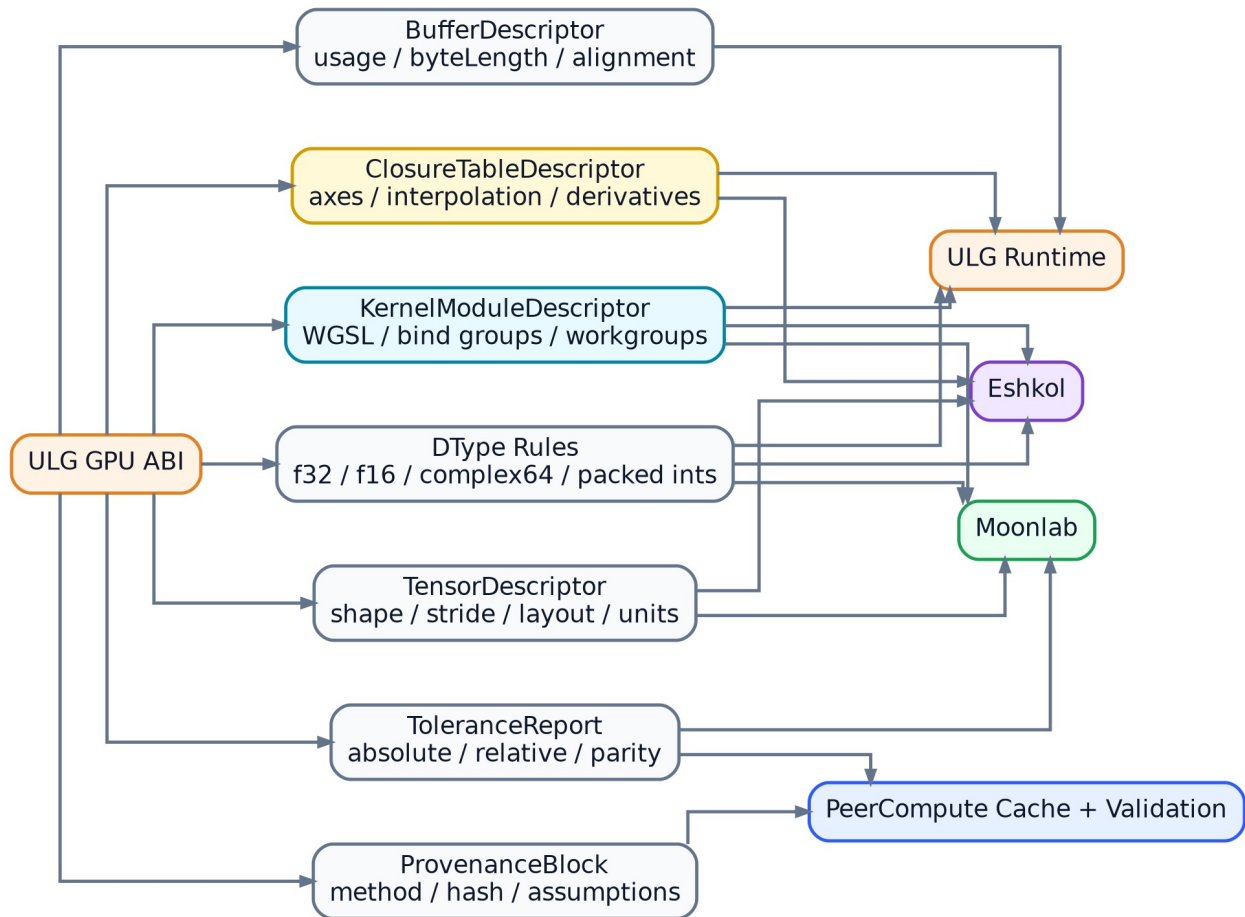


Figure 8: Shared ULG GPU ABI

The ABI should be packaged as a small shared module, not copied independently into all three repos. Recommended package boundary:

```

ulg-gpu-abi/
  src/types.ts
  src/schemas/
  src/wgsl/
  src/layouts/
  src/validation/
  conformance/
  
```

3.10 9. Data lifecycle

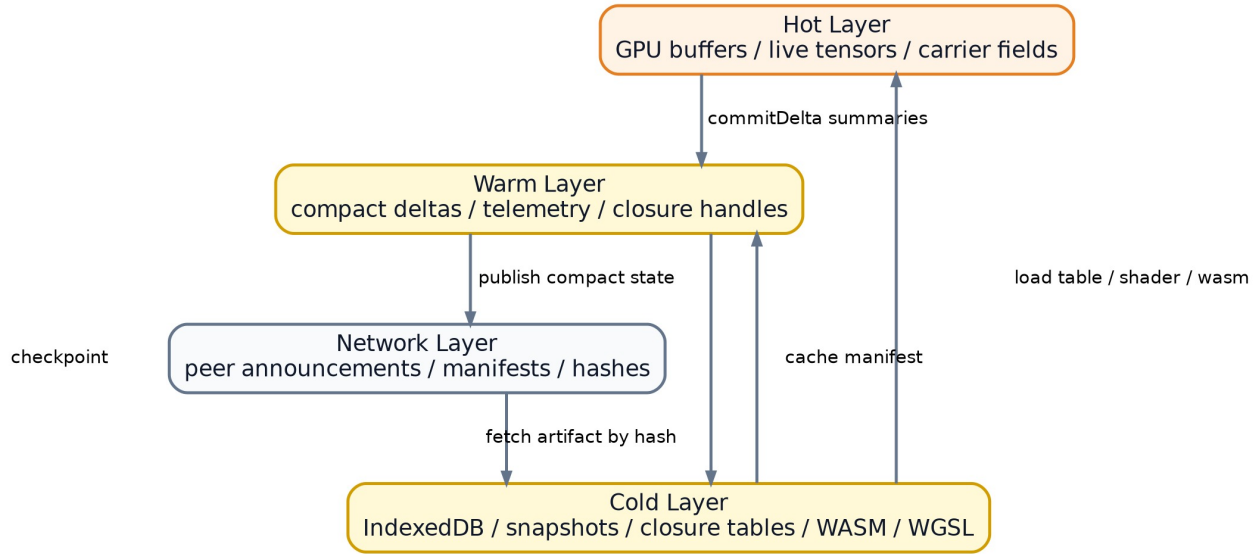


Figure 9: Data lifecycle across hot, warm, cold, and network layers

3.10.1 Hot layer

- GPU buffers;
- live tensors;
- carrier fields;
- temporary reduction buffers;
- simulation domain state.

3.10.2 Warm layer

- compact deltas;
- telemetry;
- closure handles;
- task status;
- peer announcements.

3.10.3 Cold layer

- IndexedDB snapshots;
- cached closure tables;
- WASM modules;
- WGSL modules;
- quantum response artifacts;
- validation reports.

3.11 10. End-to-end execution flows

3.11.1 10.1 Closure derivation flow

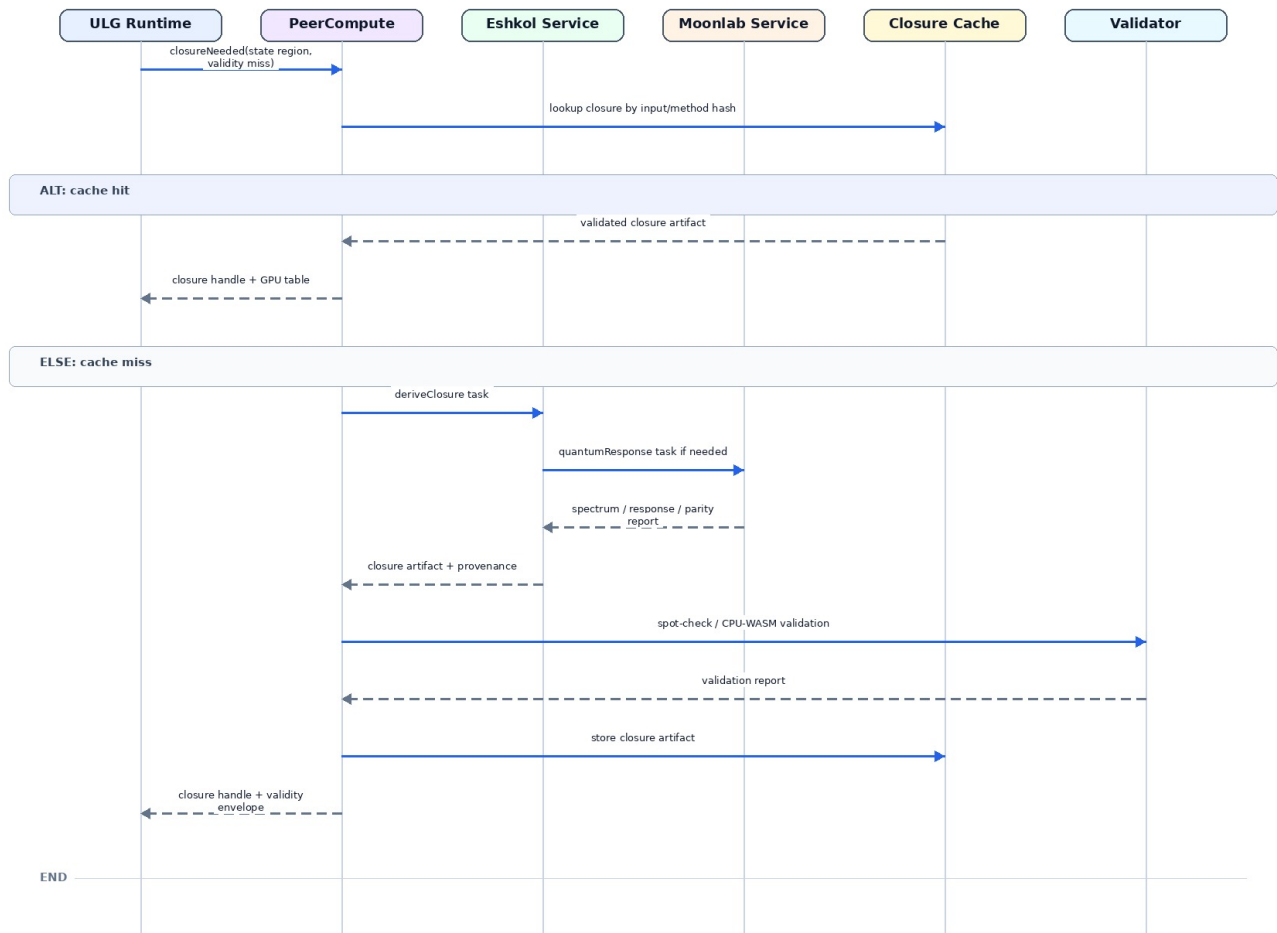


Figure 10: Closure derivation sequence

3.11.2 10.2 ULG simulation flow

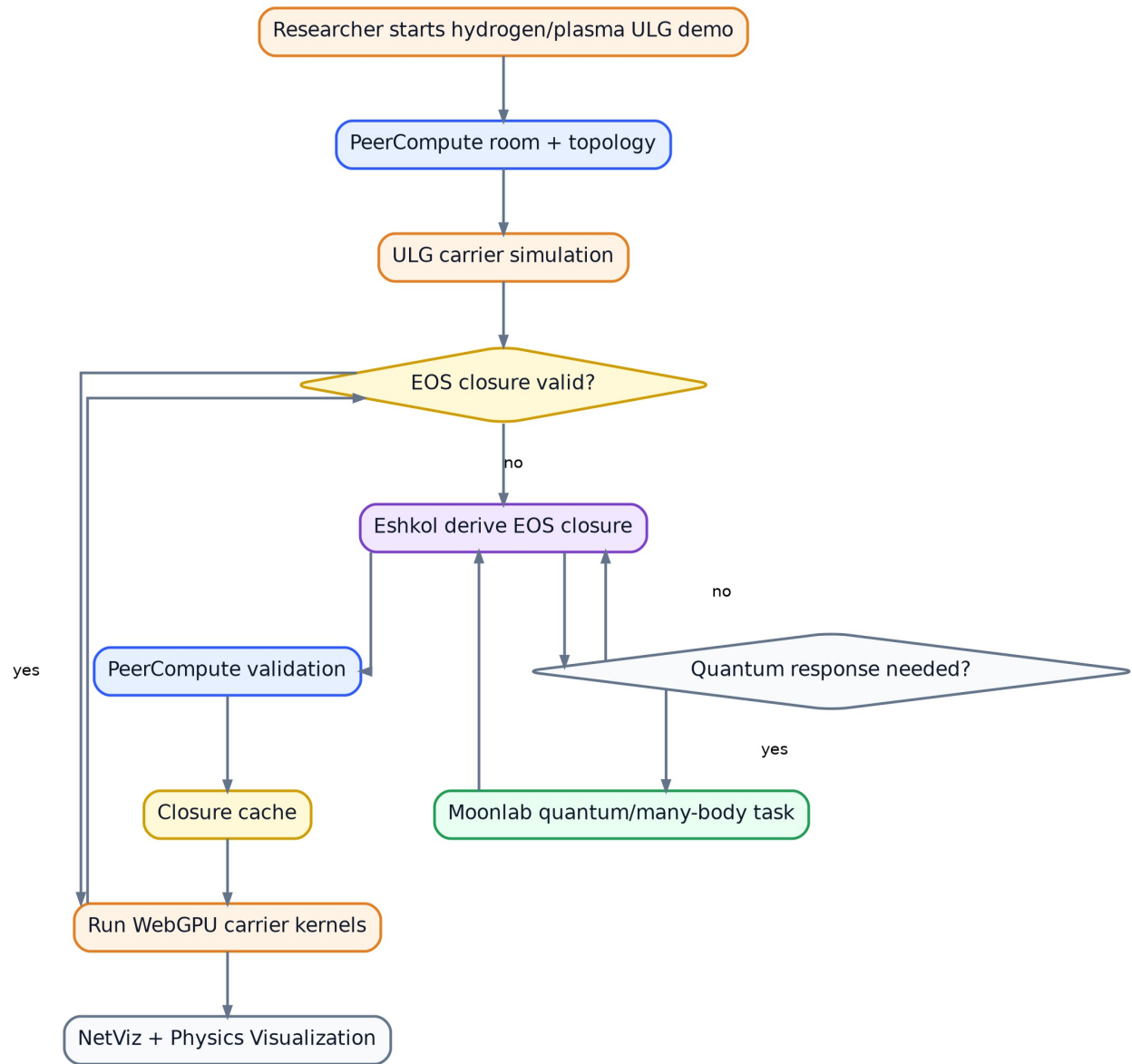


Figure 11: Hydrogen plasma demo flow

3.11.3 10.3 Task lifecycle

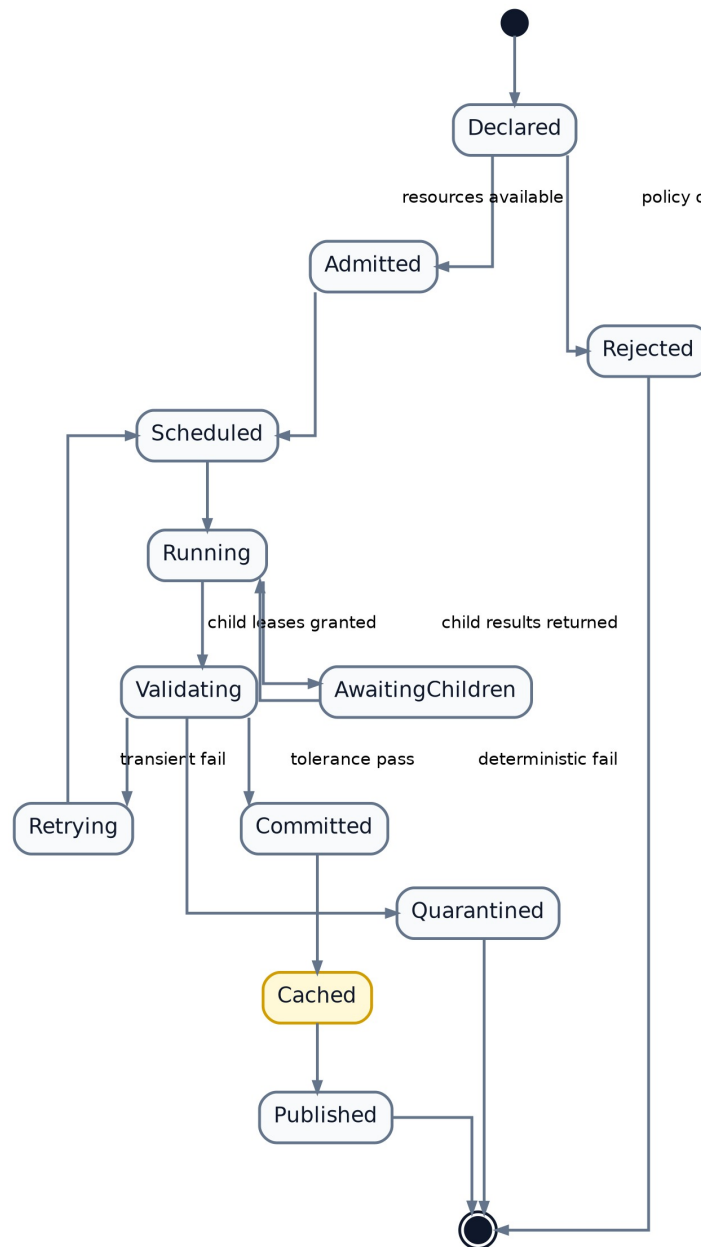


Figure 12: Task lifecycle state machine

3.12 11. Validation and provenance

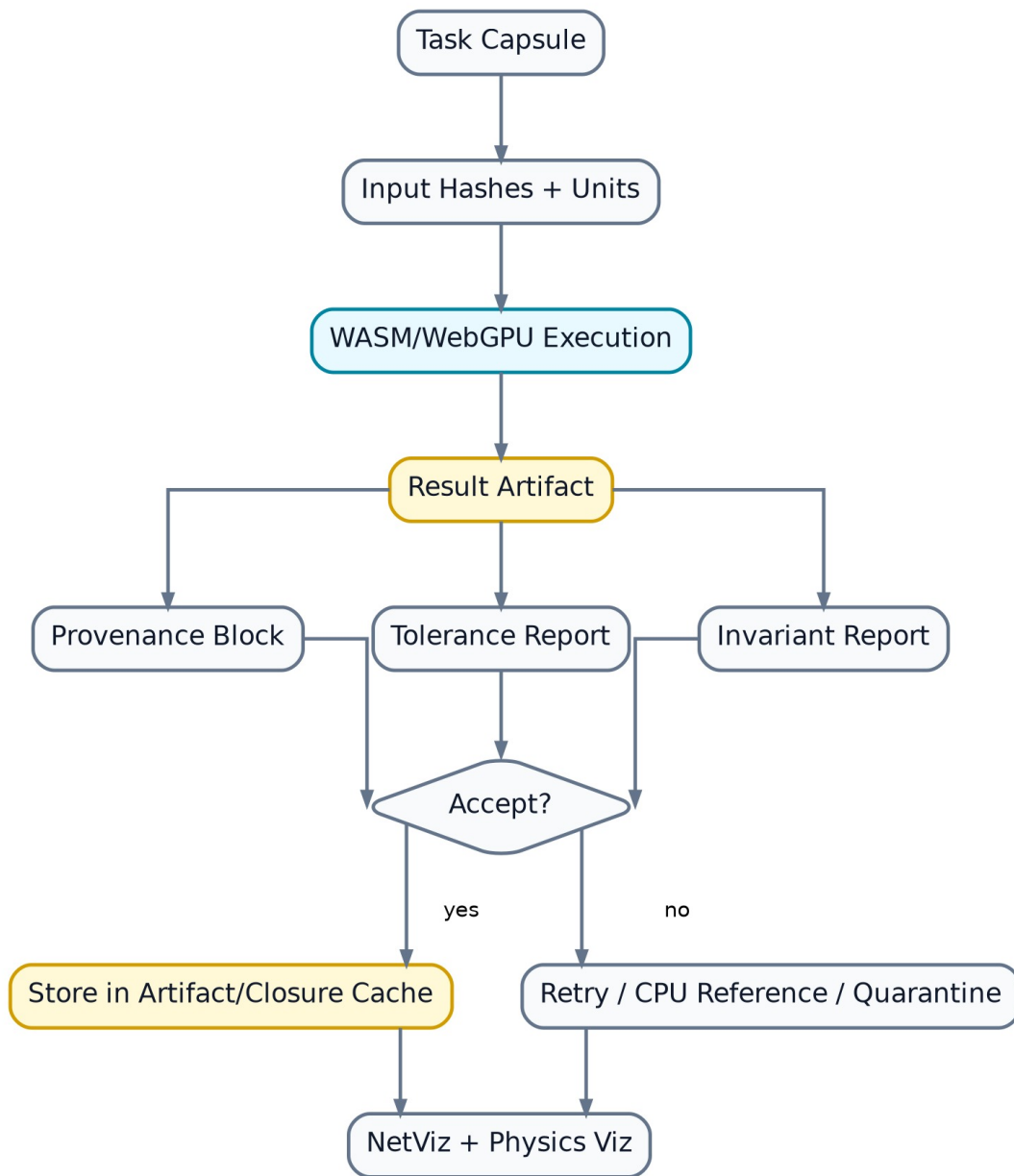


Figure 13: Validation and provenance loop

Validation layers:

1. local self-check in the service;
2. CPU/WASM reference comparison;
3. randomized spot-checks;
4. peer quorum for expensive results;
5. physics invariant audit;
6. cache quarantine on deterministic failure.

3.13 12. Visualization architecture

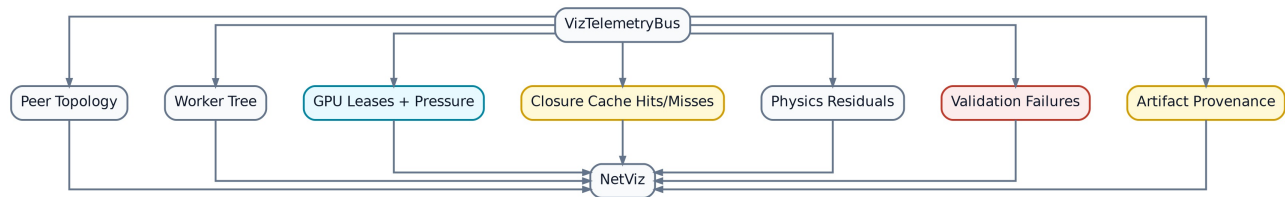


Figure 14: Visualization telemetry bus

NetViz should gain panels for:

- service registry and capabilities;
- worker tree;
- child-worker leases;
- GPU pressure;
- artifact cache;
- closure provenance;
- validation failures;
- physics residuals;
- simulation domain ownership.

3.14 13. Architecture risks

Risk	Mitigation
Eshkol and Moonlab invent incompatible GPU layouts	Shared ULG GPU ABI first
Child workers run away	Lease manager, quotas, cancellation tree
GPU compute starves visualization	GPU broker priority classes
Closure cache stores bad science	provenance + validation + quarantine
WebGPU parity differs by adapter	tolerance profiles + CPU/WASM reference
Large tensors are cloned repeatedly	transferables, SharedArrayBuffer when isolated, artifact refs
Peer network is flooded with raw state	compact deltas and content-addressed artifacts

3.15 Source notes

This specification is based on the public project documentation checked on 2026-06-05:

- [PeerCompute repository and README](#)
- [Eshkol repository and README](#)
- [Moonlab repository and README](#)
- [Moonlab architecture notes](#)
- [Moonlab WebGPU plan](#)
- [Web Workers documentation](#)
- [Worker WebGPU entry point](#)
- [SharedArrayBuffer isolation requirements](#)
- [WGSL specification](#)

4 Part 3 — Implementation Roadmap

Version: **0.5**

Date: **2026-06-05**

4.1 Abstract

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4.2 1. Roadmap philosophy

The implementation should not start by building a full multiscale physics engine. It should start by building the **contracts and orchestration substrate** that make the physics stack credible:

```
shared ABI -> supervised workers -> closure artifacts -> quantum artifacts -> validation
↪ -> end-to-end demo
```

Every milestone should leave the system runnable, testable, and observable.

4.3 2. Roadmap overview

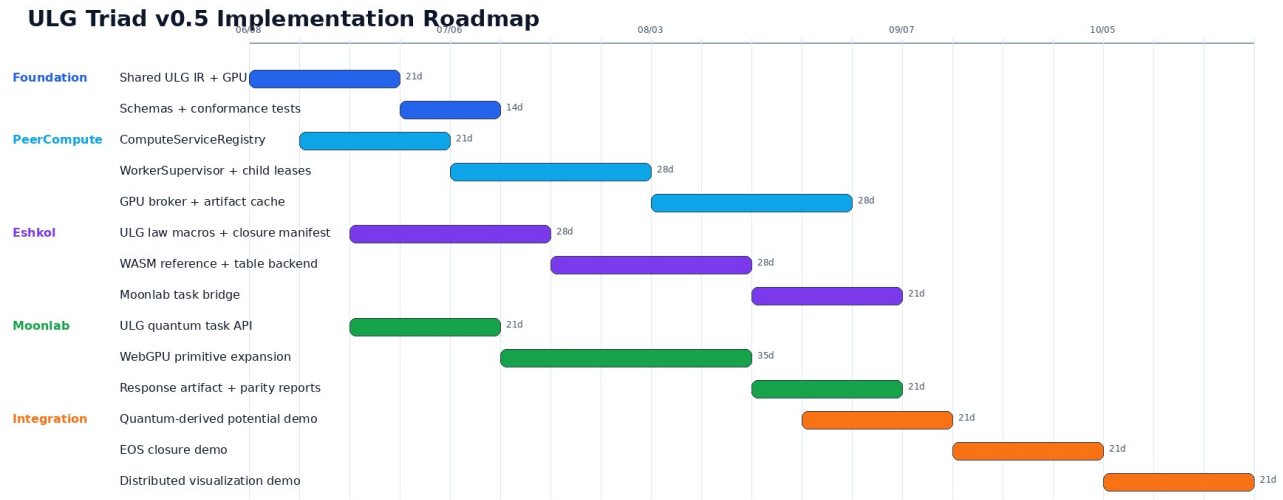


Figure 15: ULG Triad v0.5 Implementation Roadmap

4.4 3. Milestone 0.5 — specification package

Status: this document.

Deliverables:

- versioned integrated system spec;
- architecture plan;
- implementation roadmap;
- library extension plan;
- Mermaid diagrams;
- schema sketches;
- abstract for unfamiliar readers.

Acceptance:

- all documents versioned v0.5;
- all major docs include the abstract;
- diagrams compile as standalone Mermaid source;
- extension points are explicit for all three libraries.

4.5 4. Milestone 0.6 — shared ULG IR and GPU ABI

Goal: create the common contract before modifying runtime behavior deeply.

Deliverables:

- `ulg-gpu-abi` package;
- TypeScript type definitions;
- JSON schemas;
- WGSL layout snippets;
- `complex64` and tensor descriptor tests;
- closure table descriptor tests;
- tolerance report schema;
- provenance block schema.

Acceptance tests:

1. Eshkol, Moonlab, and a dummy ULG worker can all import the same ABI package.

2. A complex64 vector buffer round-trips through JS/WASM/WGSL layout tests.
3. A closure table descriptor validates against schema.
4. A tolerance report can be emitted and parsed by PeerCompute.

4.6 5. Milestone 0.7 — PeerCompute service orchestration

Goal: make PeerCompute a supervised service host.

PeerCompute work:

- ComputeServiceRegistry;
- WorkerSupervisor;
- root service worker lifecycle;
- child-worker lease manager;
- task lineage tracker;
- cancellation tree;
- service heartbeat protocol;
- minimal NetViz worker-tree panel.

Acceptance tests:

1. Register a dummy Eshkol service.
2. Register a dummy Moonlab service.
3. Submit tasks to each service.
4. Grant child-worker leases.
5. Cancel a root task and verify all child workers stop.
6. Show worker tree in NetViz.

4.7 6. Milestone 0.8 — PeerCompute GPU broker and artifact cache

Goal: broker local GPU access and store content-addressed artifacts.

PeerCompute work:

- GpuBroker capability probing;
- GPU lease request/release;
- priority classes;
- device-lost recovery path;
- ArtifactCache;
- ClosureRegistry;
- IndexedDB storage backend;
- cache announcement protocol.

Acceptance tests:

1. Service requests optional GPU lease and receives one if available.
2. Render-priority lease preempts background compute lease.
3. Device-lost event marks running tasks retryable.
4. Artifact cache stores and retrieves by content hash.
5. Closure registry invalidates a closure when validity conditions fail.

4.8 7. Milestone 0.9 — Eshkol closure service

Goal: run Eshkol as a supervised closure compiler.

Eshkol work:

- service worker bootstrap;
- ULG law macro prototype;
- closure manifest emitter;
- derivative metadata export;
- WASM reference function export;
- closure table generation over tiles;

- child-worker tiling;
- optional WGSL/table strategy descriptor;
- PeerCompute task protocol adapter.

Acceptance tests:

1. Eshkol service compiles a simple free-energy closure.
2. Eshkol emits ClosureArtifact with validity and provenance.
3. Eshkol generates a closure table tile using child workers.
4. Eshkol emits derivative metadata for pressure/entropy-style outputs.
5. PeerCompute validates random table samples against WASM reference.

4.9 8. Milestone 1.0 — Moonlab quantum service

Goal: run Moonlab as a supervised quantum/many-body response backend.

Moonlab work:

- service worker bootstrap;
- ULG quantum task API;
- WebGPU capability probe;
- CPU fallback path;
- response artifact exporter;
- parity report emitter;
- WebGPU primitive expansion for complex vector ops, reductions, sparse matvec, and tensor-network kernels;
- PeerCompute task protocol adapter.

Acceptance tests:

1. Moonlab service runs a small CPU reference quantum task.
2. Moonlab runs an equivalent WebGPU task when available.
3. Moonlab emits QuantumResponseArtifact.
4. Moonlab emits CPU/GPU parity report.
5. Failed parity quarantines the artifact instead of caching it as valid.

4.10 9. Milestone 1.1 — Moonlab → Eshkol closure pipeline

Goal: connect quantum response artifacts to closure derivation.

End-to-end demo 1: quantum-derived bond potential.

Flow:

```
Moonlab computes energy samples -> Eshkol fits/interpolates E(r) -> Eshkol differentiates
↪ force -> ULG worker consumes closure table
```

Acceptance tests:

1. Force equals $-dE/dr$ within tolerance.
2. Closure invalidates outside sampled range.
3. CPU/WASM reference and WGSL interpolation agree within tolerance.
4. PeerCompute displays artifact provenance.

4.11 10. Milestone 1.2 — EOS closure pipeline

Goal: demonstrate ULG's material/plasma closure factory.

End-to-end demo 2: hydrogen/plasma EOS closure.

Flow:

```
Eshkol defines free-energy model -> derives pressure/entropy/heat-capacity/ionization
↪ outputs -> PeerCompute validates -> ULG carrier kernel interpolates closure
```

Acceptance tests:

1. Closure artifact includes inputs rho, T, species.
2. Outputs include P, S, Cv, and ionization-like variables.
3. Derivatives $dP/d\rho$ and dP/dT are available or explicitly unavailable.
4. Carrier simulation can use the closure without pulling full Eshkol into the hot kernel.

4.12 11. Milestone 1.3 — distributed visualization and telemetry

Goal: make the system debuggable.

PeerCompute/NetViz work:

- worker-tree panel;
- GPU pressure panel;
- closure cache panel;
- validation/provenance panel;
- task lineage panel;
- physics residual overlay;
- domain ownership view.

Acceptance tests:

1. NetViz shows root workers and child workers.
2. NetViz shows GPU leases by service.
3. NetViz shows closure cache hits/misses.
4. NetViz can trace a ULG closure from simulation use back to Eshkol and Moonlab artifacts.

4.13 12. Milestone 1.4 — hardening

Work:

- fuzz schemas;
- soak-test worker leaks;
- stress-test child-worker quotas;
- simulate WebGPU device loss;
- simulate peer churn;
- validate cache quarantine behavior;
- benchmark table interpolation;
- benchmark Moonlab WebGPU primitives;
- benchmark Eshkol closure generation.

Acceptance:

- no runaway worker trees;
- no untracked GPU allocations;
- no uncancellable long-running tasks;
- no artifact accepted without validation status;
- degraded browsers fall back to WASM/CPU where possible.

4.14 13. First three demos

4.14.1 Demo A — Supervised service smoke test

PeerCompute starts Eshkol + Moonlab services -> both spawn child workers -> both report
 ↪ telemetry -> PeerCompute cancels both cleanly

4.14.2 Demo B — Quantum-derived potential

Moonlab energy samples -> Eshkol closure -> ULG two-particle oscillator -> NetViz
 ↪ provenance

4.14.3 Demo C — Hydrogen/plasma closure

Eshkol EOS table -> ULG carrier collapse toy model -> closure invalidation and refresh

4.15 14. Definition of done for v1-compatible integration

The integration is credible when:

- PeerCompute supervises all root and child workers;
- Moonlab and Eshkol never publish distributed state directly;
- all heavy artifacts are content-addressed;
- WebGPU tasks have CPU/WASM fallback or explicit unsupported status;
- closure artifacts include validity, uncertainty, and provenance;
- NetViz shows enough information to debug a failed closure or quantum task;
- the first two end-to-end demos run in a browser peer.

4.16 Source notes

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- [Web Workers documentation](#)
- [Worker WebGPU entry point](#)
- [SharedArrayBuffer isolation requirements](#)
- [WGSL specification](#)

5 Part 4 — Library Extension Plan

Version: **0.5**

Date: **2026-06-05**

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5.2 1. Extension strategy

All three libraries should be extended, but not equally. PeerCompute needs orchestration and supervision primitives. Eshkol needs ULG law/closure authoring and compilation outputs. Moonlab needs ULG quantum-task and response-artifact surfaces. The shared ULG GPU ABI should be built before the libraries grow separate incompatible GPU data layouts.

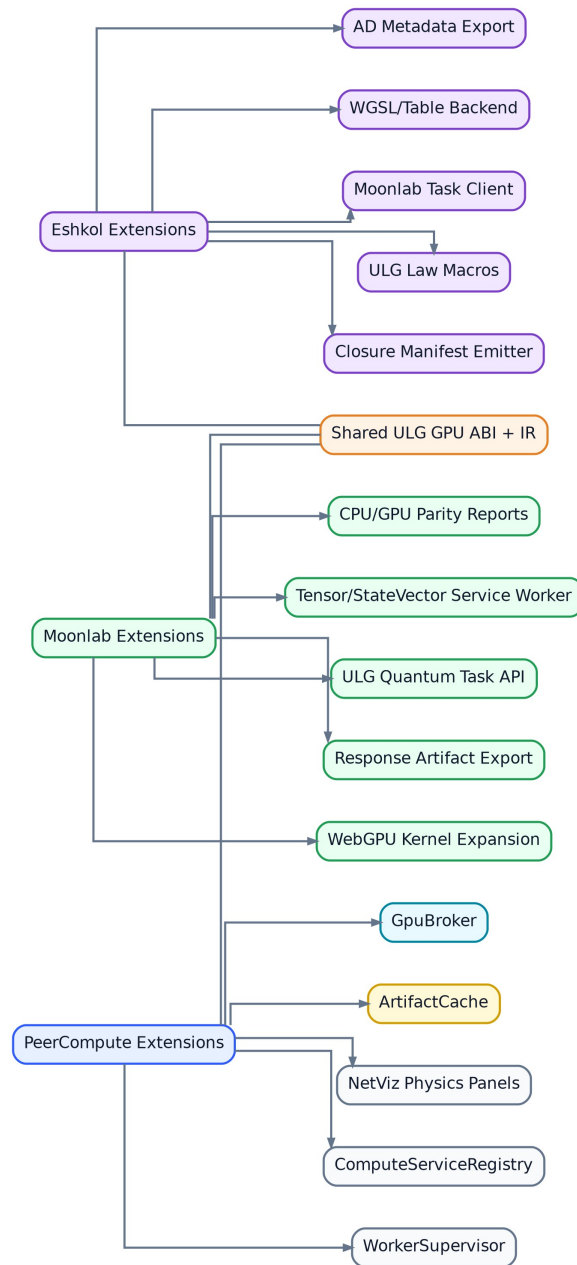


Figure 16: Library extension map

5.3 2. PeerCompute extension plan

5.3.1 2.1 New packages/modules

Recommended modules:

```
peercompute/compute-services/
  ComputeServiceRegistry.ts
  ComputeServiceManifest.ts
  ServiceWorkerHost.ts
```

```
peercompute/worker-supervisor/
  WorkerSupervisor.ts
```

```

ChildWorkerLeaseManager.ts
WorkerTreeTelemetry.ts
CancellationTree.ts

peercompute/gpu-broker/
  GpuBroker.ts
  GpuLease.ts
  GpuCapabilityProbe.ts
  DeviceLostRecovery.ts

peercompute/artifacts/
  ArtifactCache.ts
  ClosureRegistry.ts
  ProvenanceIndex.ts
  ValidationStore.ts

peercompute/netviz-panels/
  WorkerTreePanel.tsx
  GpuPressurePanel.tsx
  ClosureCachePanel.tsx
  ProvenancePanel.tsx
  PhysicsResidualPanel.tsx

```

5.3.2 2.2 ComputeServiceRegistry

Add service registration for long-lived compute services.

```

await node.compute.registerService({
  serviceId: "eshkol",
  runtime: "wasm-webgpu",
  entry: { workerModule: "/workers/eshkol-service.worker.js" },
  childWorkers: { allowed: true, maxChildren: 8, allowedModules:
    ↪ ["/workers/eshkol-child.worker.js"], sameOriginOnly: true },
  capabilities: ["ulg.closure.derive", "ulg.ad", "ulg.table.generate"],
  abi: { ulgIrVersion: "0.5", gpuAbiVersion: "0.5", supportedDTypes: ["f32",
    ↪ "complex64"], supportedLayouts: ["soa", "interleaved"] },
  validation: { requiresCpuReference: true, toleranceProfile: "scientific-default",
    ↪ parityModes: ["wasm-reference"] }
})

```

5.3.3 2.3 Worker supervision

Add:

- root worker lifecycle;
- child lease request handling;
- heartbeat monitoring;
- cancellation tree;
- worker crash recovery;
- worker tree telemetry for NetViz.

5.3.4 2.4 GPU broker

Add:

- adapter/capability probing;
- lease priority;
- memory pressure estimates;

- render-vs-compute scheduling policy;
- device-lost handling;
- timing hooks;
- WebGPU unsupported fallback.

5.3.5 2.5 Artifact and closure cache

Add:

- content-addressed storage;
- IndexedDB backend;
- closure validity queries;
- invalidation events;
- quarantine status;
- peer cache announcements;
- provenance lookup.

5.3.6 2.6 NetViz extensions

NetViz should visualize:

- active services;
- worker tree;
- child-worker leases;
- GPU leases;
- task lineage;
- artifact provenance;
- closure cache hits/misses;
- validation failures;
- physics residuals.

5.4 3. Eshkol extension plan

5.4.1 3.1 New capabilities

Eshkol should add:

```

ULG law macros
ULG closure manifest emitter
unit/dimensional metadata export
validity-envelope declarations
provenance block emission
Moonlab task client
WASM reference export
WGSL kernel/table strategy backend
closure-table tiling
PeerCompute service worker adapter

```

5.4.2 3.2 Example Eshkol ULG closure shape

```

(define-ulg-closure eos-hydrogen-plasma
  (inputs
    (rho density)
    (T temperature)
    (species species-vector))
  (outputs
    (P pressure)
    (S entropy)

```

```

    (Cv heat-capacity)
    (ion ionization-fraction))
  (assumptions
    local-thermodynamic-equilibrium
    hydrogen-helium-plasma)
  (validity
    (temperature 1e3 1e8)
    (density 1e-12 1e6))
  (body
    (let ((F (+ (ideal-ion-free-energy rho T species)
                (fermi-electron-free-energy rho T species)
                (radiation-free-energy T))))
      (values
        (- (d/d-volume F))
        (- (d/d-temperature F))
        (heat-capacity-from F T)
        (ionization-equilibrium rho T species))))))

```

5.4.3 3.3 Eshkol outputs

For every closure, Eshkol should emit:

1. ClosureArtifact manifest;
2. WASM reference function artifact;
3. derivative metadata;
4. optional WGS� module or table-interpolation strategy;
5. validity envelope;
6. uncertainty and validation hints;
7. provenance block;
8. optional Moonlab dependency list.

5.4.4 3.4 Eshkol service tasks

```

eshkol.law.compile
eshkol.closure.derive
eshkol.closure.table.generate
eshkol.derivative.export
eshkol.wasm.reference.build
eshkol.wgsl.emit
eshkol.validation.reference

```

5.4.5 3.5 Eshkol acceptance tests

- Compile a ULG closure into a manifest.
- Export a WASM reference function.
- Generate a tiled closure table with child workers.
- Validate random table samples against WASM reference.
- Emit derivative metadata.
- Request a Moonlab quantum-response task through PeerCompute.

5.5 4. Moonlab extension plan

5.5.1 4.1 New capabilities

Moonlab should add:

```

ULG quantum task API
ULG response artifact exporter
service worker bootstrap
WebGPU capability report in service manifest
CPU/WebGPU parity report
complex64 tensor ABI support
sparse Hamiltonian matvec task
Lanczos/Krylov task
state-vector and tensor-network task adapters
transition/correlation response tasks

```

5.5.2 4.2 Moonlab service tasks

```

moonlab.quantum.response
moonlab.quantum.spectrum
moonlab.quantum.ground_state
moonlab.quantum.transition_matrix
moonlab.quantum.correlation
moonlab.tensor_network.contract
moonlab.tensor_network.expectation
moonlab.parity.cpu_webgpu

```

5.5.3 4.3 Moonlab artifact shape

```

{
  "artifactId": "quantum.response.local_patch.001",
  "sourceService": "moonlab",
  "taskKind": "quantum.response",
  "method": "lanczos-demo",
  "representation": "state_vector",
  "outputs": {
    "energyLevels": "artifact://...",
    "transitionMatrixElements": "artifact://...",
    "forceSamples": "artifact://..."
  },
  "uncertainty": {
    "truncationError": 0.0,
    "parityError": 0.0
  },
  "validation": {
    "status": "pass",
    "validationMode": "cpu-reference"
  }
}

```

5.5.4 4.4 Moonlab WebGPU priorities

Prioritize kernels that directly help ULG closure generation:

1. complex vector operations;
2. complex dot/norm reductions;
3. sparse Hamiltonian matvec;
4. Krylov/Lanczos loop primitives;
5. tensor contraction primitives;
6. MPS two-site updates;
7. expectation-value reductions;

- 8. probability reductions;
- 9. CPU/GPU parity harness.

5.5.5 4.5 Moonlab acceptance tests

- Run a small CPU quantum task.
- Run the same task on WebGPU where available.
- Emit a parity report.
- Emit a QuantumResponseArtifact.
- Quarantine failed parity outputs.
- Feed a response artifact to Eshkol closure derivation.

5.6 5. Shared ULG GPU ABI package

Create a shared package used by all three libraries.

```
ulg-gpu-abi/
  package.json
  src/types.ts
  src/layouts/complex64.ts
  src/layouts/tensor.ts
  src/layouts/closure-table.ts
  src/wgsl/common.wgsl
  schemas/*.json
  conformance/*.test.ts
```

5.6.1 Required descriptors

- DTypeDescriptor;
- TensorDescriptor;
- ComplexLayoutDescriptor;
- ClosureTableDescriptor;
- KernelModuleDescriptor;
- ToleranceReport;
- ProvenanceBlock.

5.7 6. Cross-repository integration plan

5.7.1 Step 1 — publish ABI package

All repos import the shared types. No GPU backend work proceeds until the ABI package exists.

5.7.2 Step 2 — dummy service integration

PeerCompute registers dummy Eshkol and Moonlab services that return mock artifacts.

5.7.3 Step 3 — real service bootstraps

Eshkol and Moonlab run as WASM service workers under PeerCompute.

5.7.4 Step 4 — real child-worker leases

Both services use leased child workers rather than unsupervised spawning.

5.7.5 Step 5 — real artifacts

Moonlab emits a quantum artifact. Eshkol emits a closure artifact. PeerCompute caches and displays both.

5.7.6 Step 6 — ULG consumes closure

A toy ULG WebGPU kernel uses the closure table and emits compact deltas.

5.8 7. Backward compatibility

- PeerCompute keeps existing compute-task APIs; service registry is additive.
- Eshkol keeps existing language/compiler behavior; ULG macros are additive.
- Moonlab keeps existing bindings; ULG task API is additive.
- WebGPU support remains optional with CPU/WASM fallback.

5.9 8. Repository ownership boundaries

Concern	Owner
Network authority	PeerCompute
Distributed scheduling	PeerCompute
Worker tree	PeerCompute
GPU leases	PeerCompute
Law language	Eshkol
AD / closure math	Eshkol
Closure artifact emission	Eshkol
Quantum state/tensor simulation	Moonlab
Quantum response artifact emission	Moonlab
Shared tensor/complex ABI	shared package
Hot carrier kernels	ULG runtime / PeerCompute service
Visualization	PeerCompute / NetViz

5.10 9. What not to build

Do not build:

- an independent Eshkol P2P scheduler;
- an independent Moonlab P2P scheduler;
- separate GPU ABIs for Eshkol and Moonlab;
- hidden child-worker spawns outside PeerCompute;
- full tensor replication through network messages;
- artifact caches without provenance;
- WebGPU-only correctness paths without CPU/WASM fallback.

5.11 Source notes

This specification is based on the public project documentation checked on 2026-06-05:

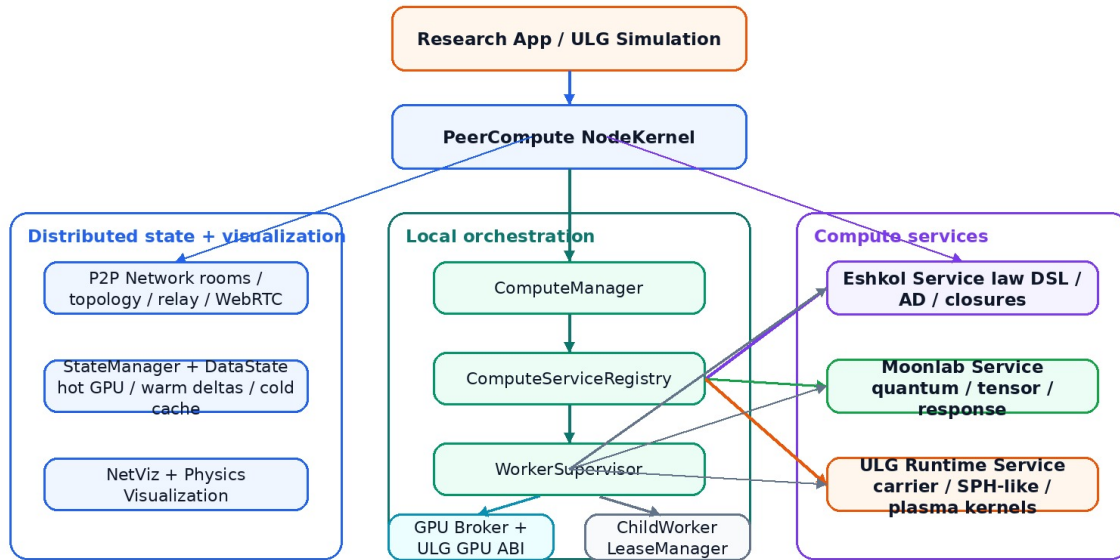
- [PeerCompute repository and README](#)
- [Eshkol repository and README](#)
- [Moonlab repository and README](#)
- [Moonlab architecture notes](#)
- [Moonlab WebGPU plan](#)
- [Web Workers documentation](#)
- [Worker WebGPU entry point](#)
- [SharedArrayBuffer isolation requirements](#)
- [WGSL specification](#)

6 Appendix A — Standalone Mermaid Diagram Sources

These are the standalone Mermaid diagram sources from the v0.5 package, embedded so the file can be rendered without external .mmd files.

6.1 01_system_context.mmd

PeerCompute + Eshkol + Moonlab: ULG Orchestration Context



Closure and simulation data loop

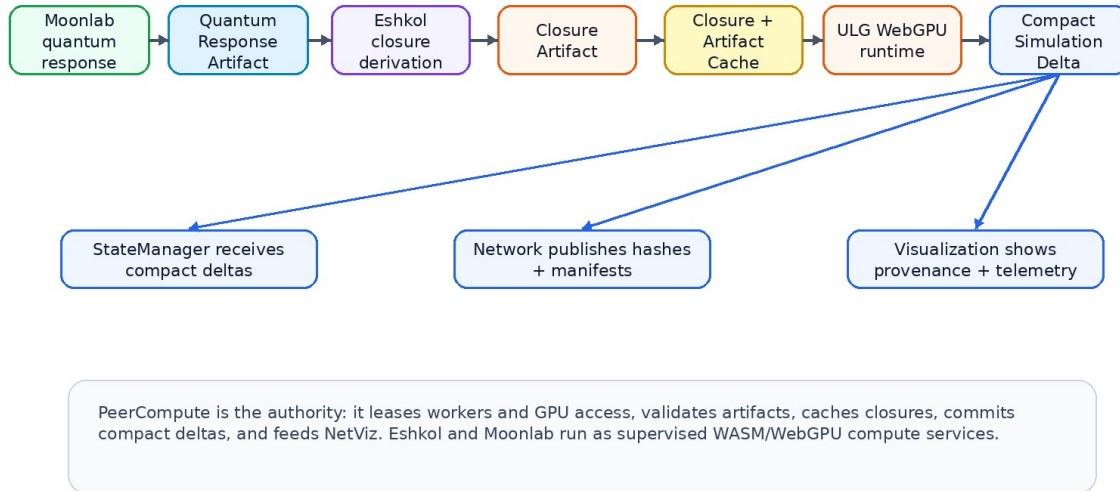


Figure 17: System context and orchestration loop

6.2 02_supervised_worker_tree.mmd

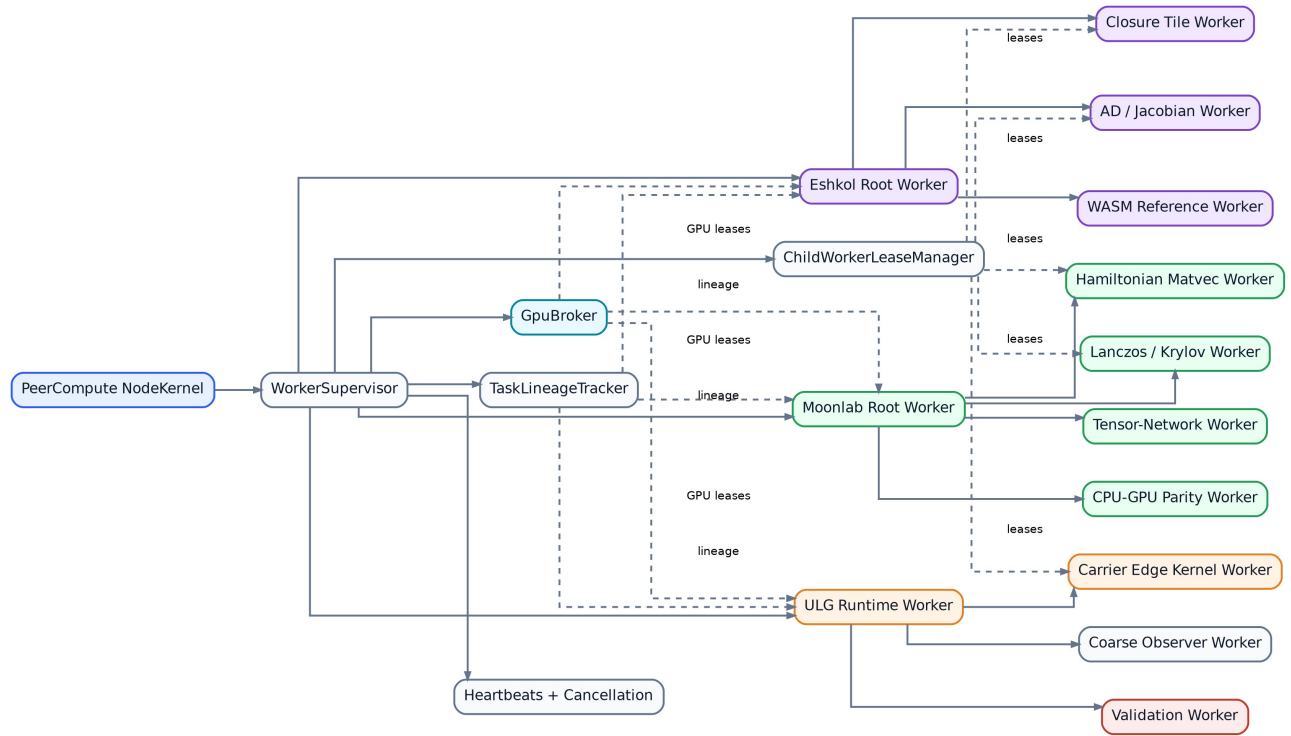


Figure 18: Supervised worker tree

6.3 03_closure_factory_flow.mmd

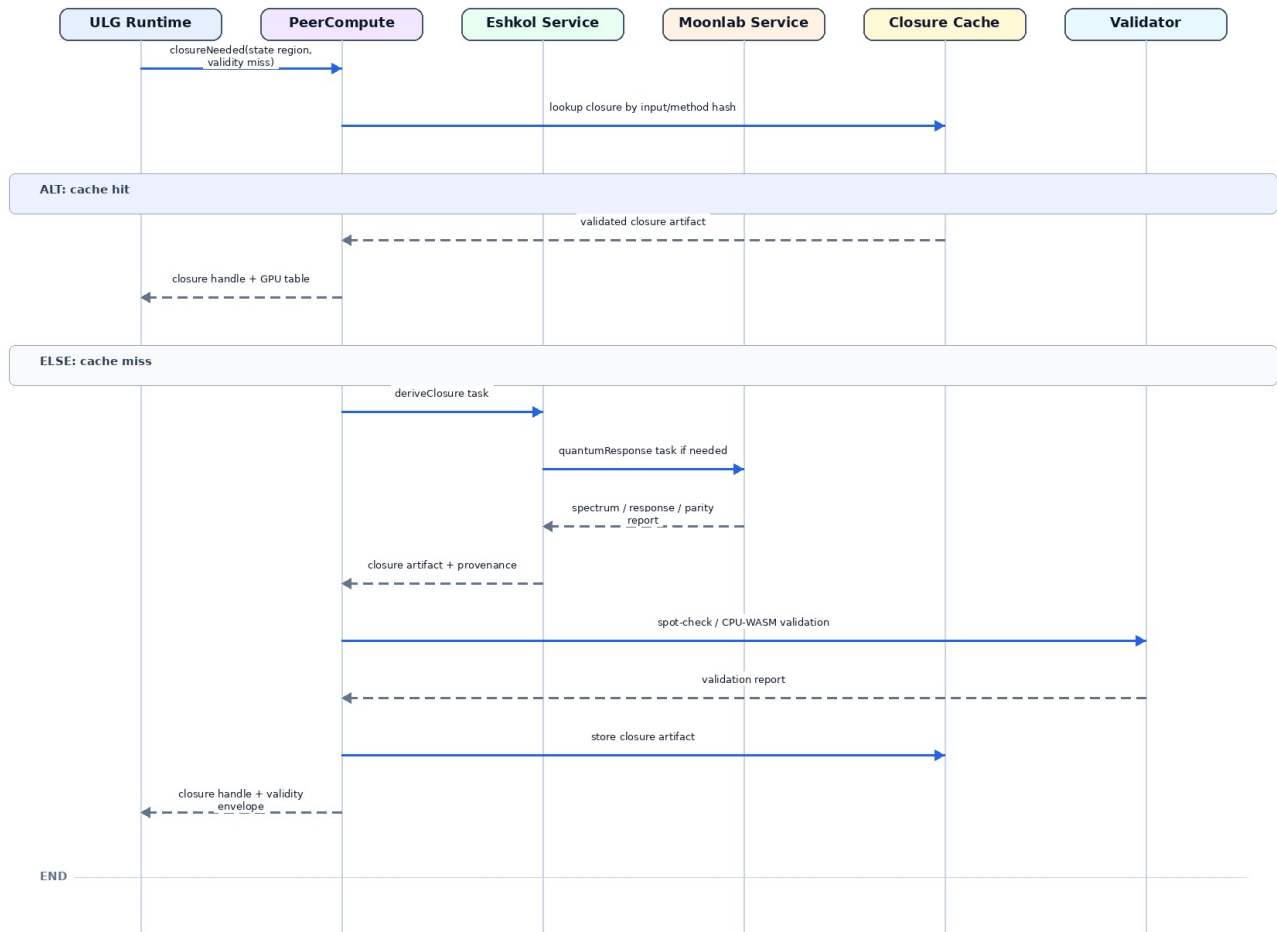


Figure 19: Closure derivation sequence

6.4 04_shared_gpu_abi.mmd

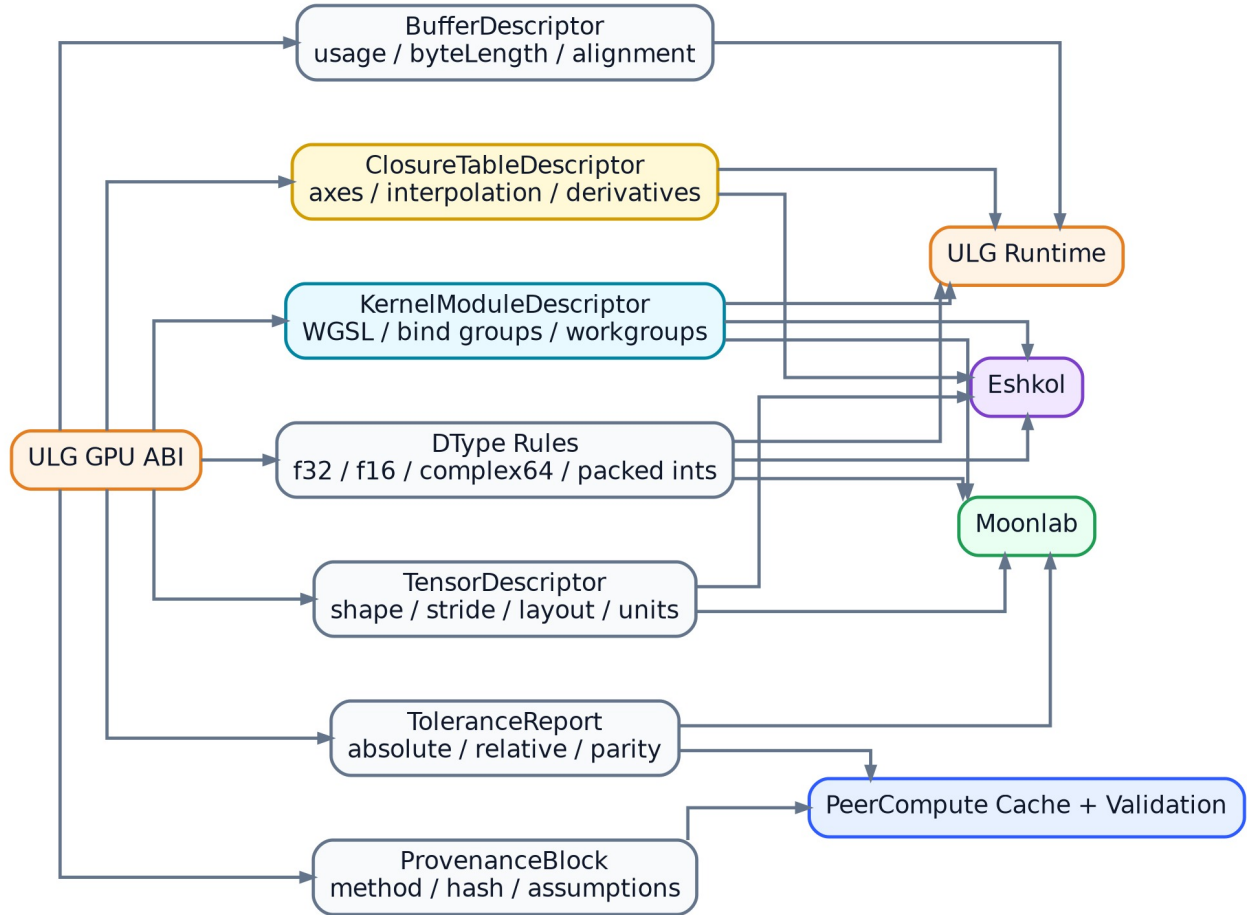


Figure 20: Shared ULG GPU ABI

6.5 05_data_lifecycle.mmd

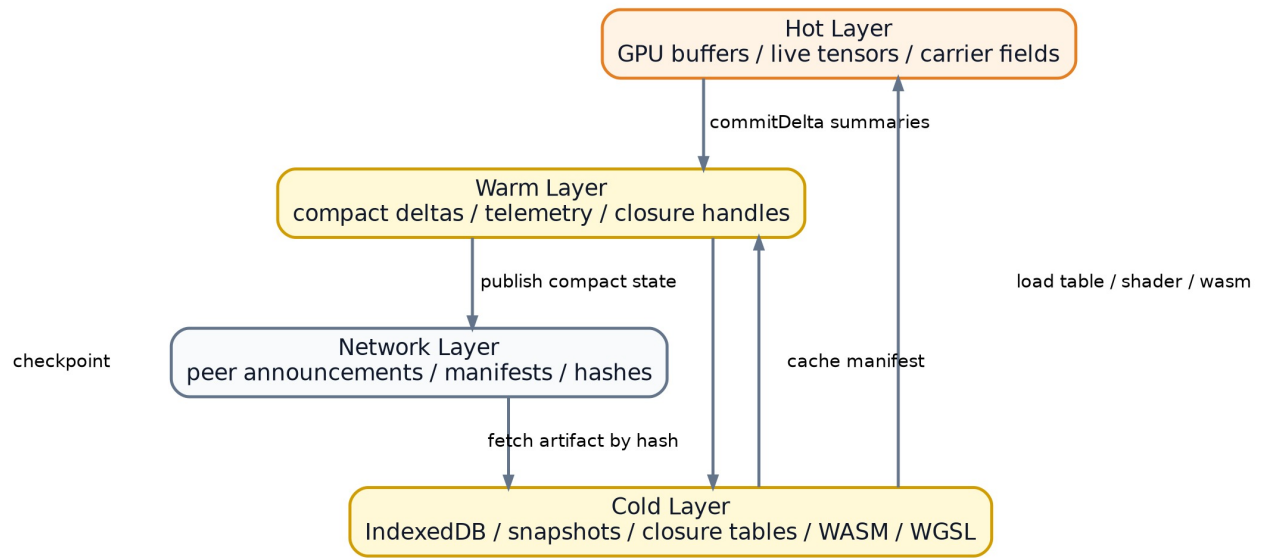


Figure 21: Data lifecycle across hot, warm, cold, and network layers

6.6 06_task_state_machine.mmd

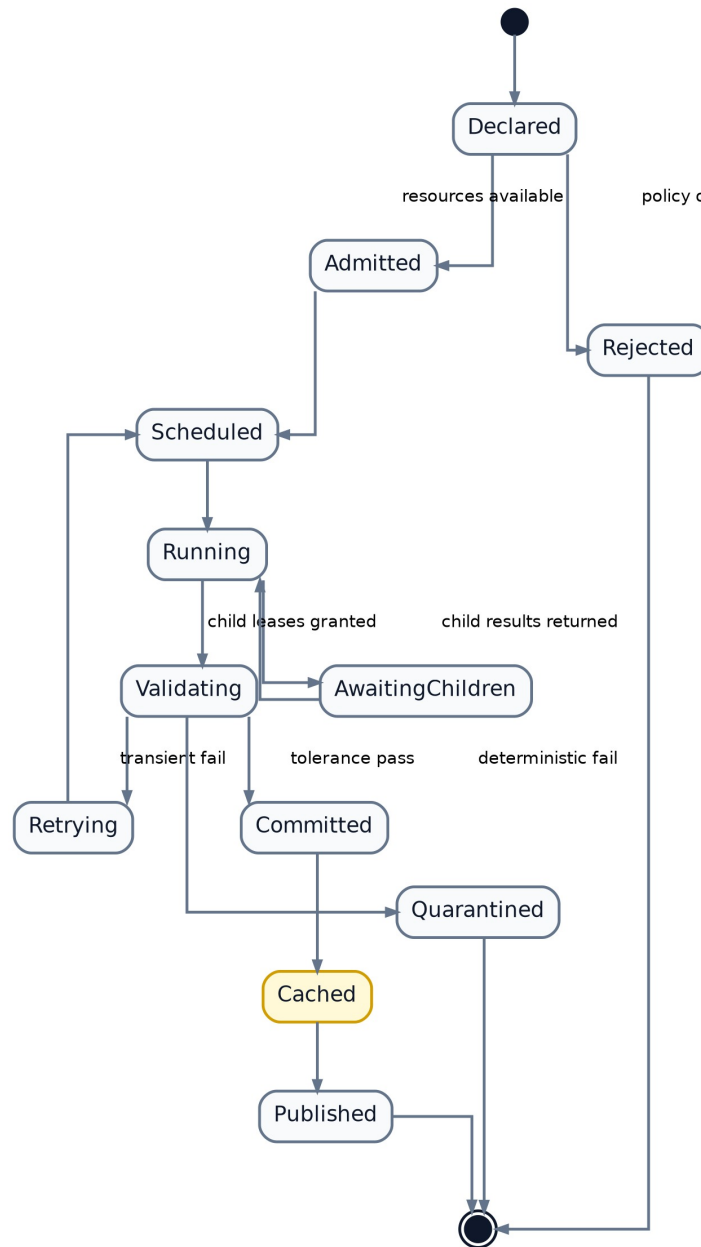


Figure 22: Task lifecycle state machine

6.7 07_validation_provenance_loop.mmd

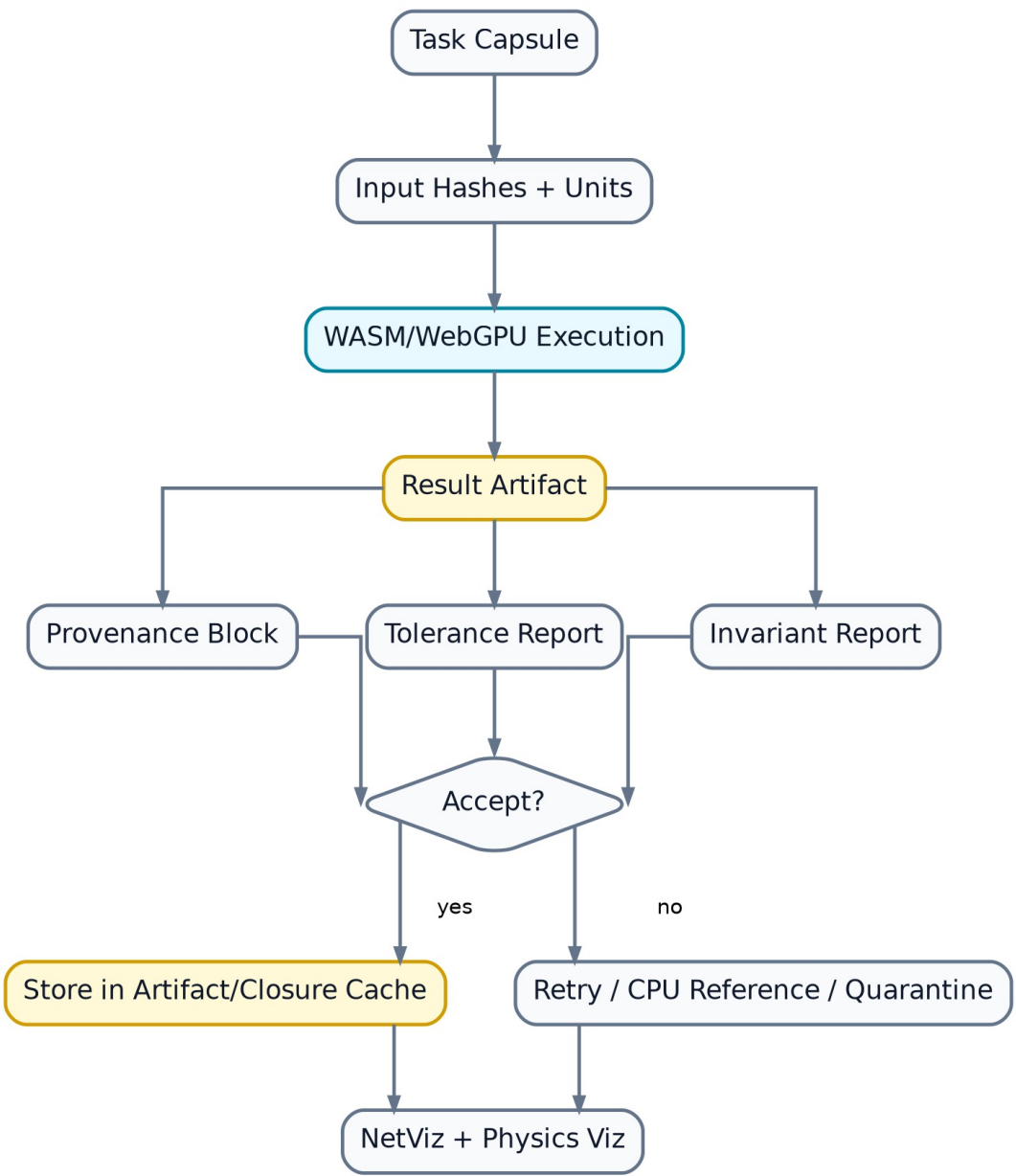


Figure 23: Validation and provenance loop

6.8 08_library_extension_map.mmd

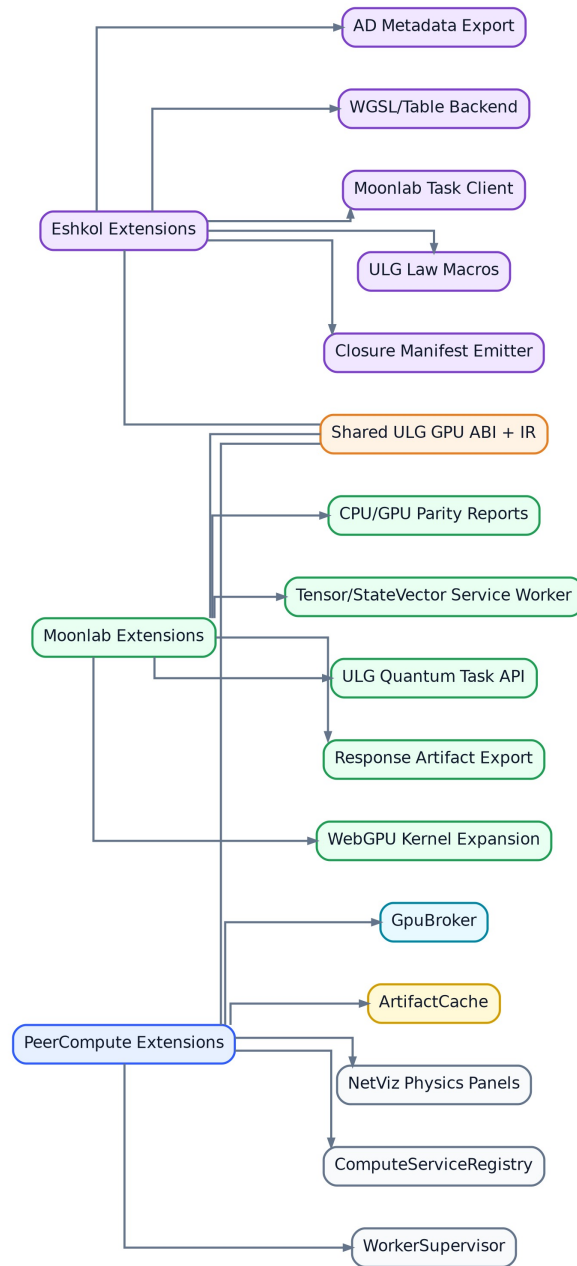


Figure 24: Library extension map

6.9 09_roadmap.mmd

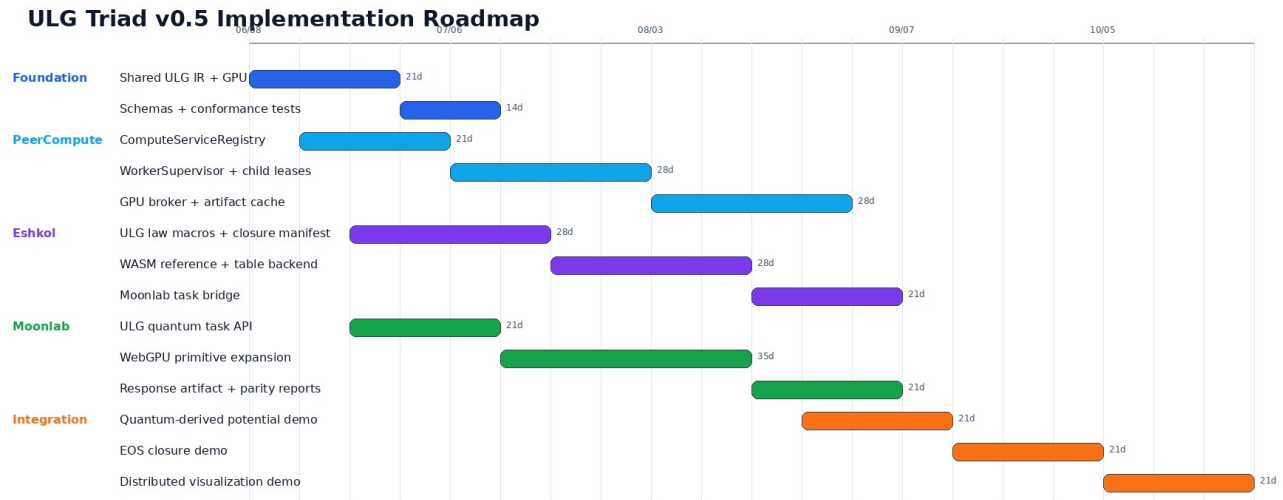


Figure 25: ULG Triad v0.5 Implementation Roadmap

6.10 10_visualization_telemetry.mmd

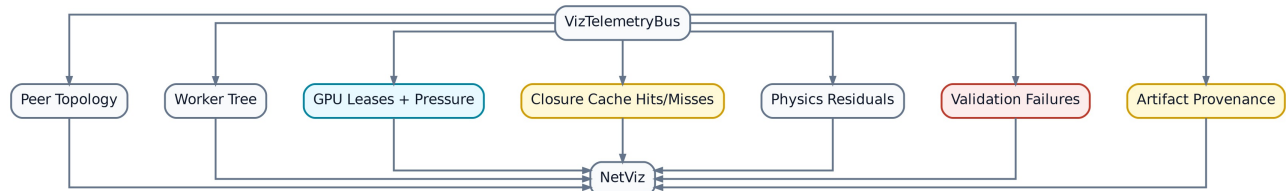


Figure 26: Visualization telemetry bus

6.11 11_security_resource_model.mmd

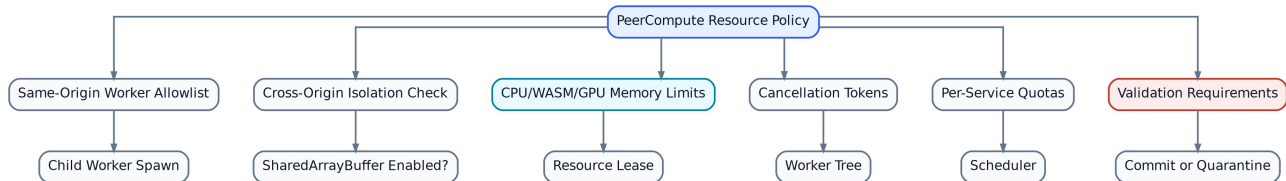


Figure 27: Resource and security policy

6.12 12_end_to_end_demo.mmd

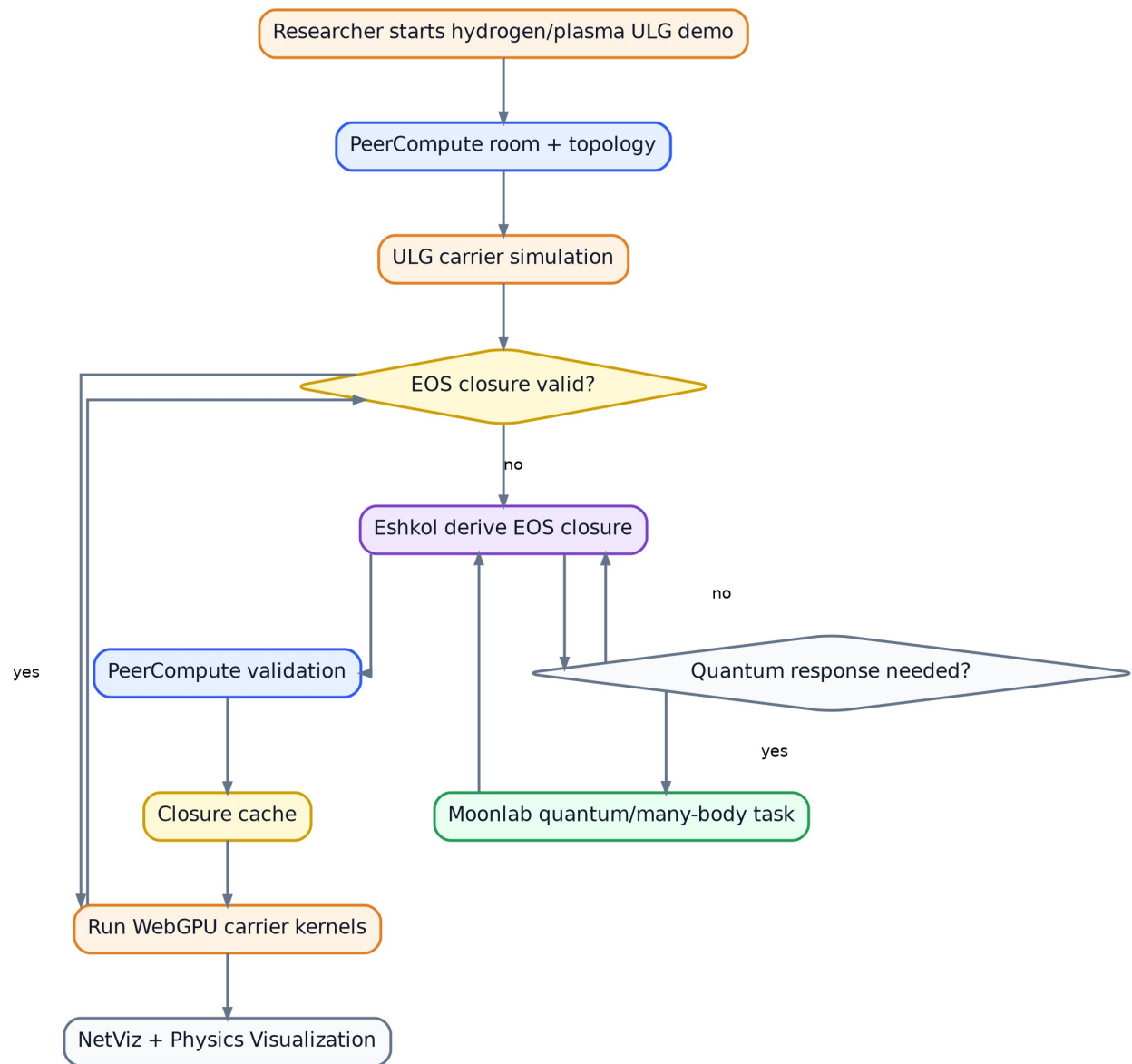


Figure 28: Hydrogen plasma demo flow

6.13 Diagram index notes

Version: **0.5**

Date: **2026-06-05**

This directory contains standalone Mermaid source diagrams used by the v0.5 Markdown documents.

6.14 01 System Context

PeerCompute + Eshkol + Moonlab: ULG Orchestration Context

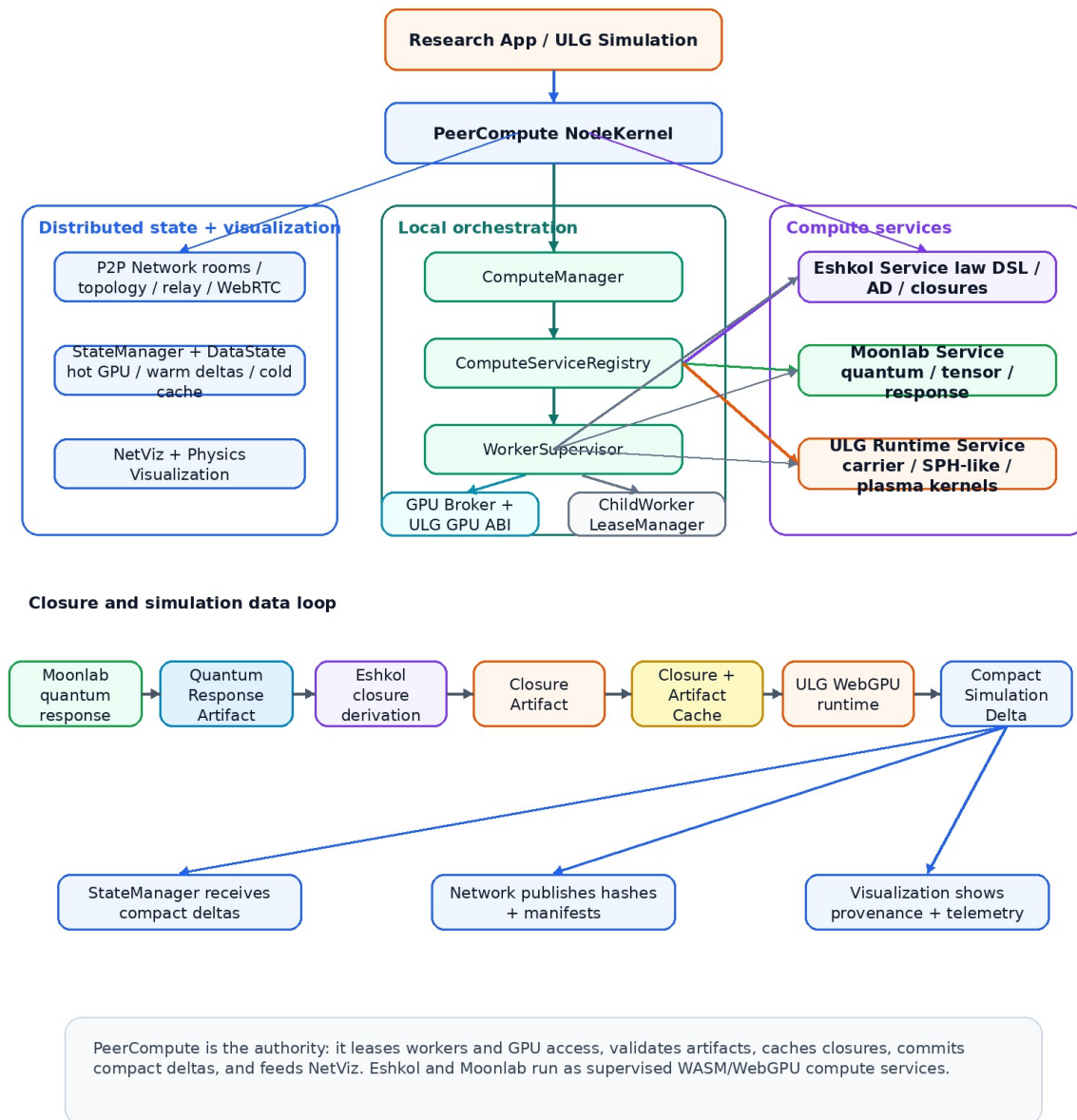


Figure 29: System context and orchestration loop

File: 01_system_context.mmd

6.15 02 Supervised Worker Tree

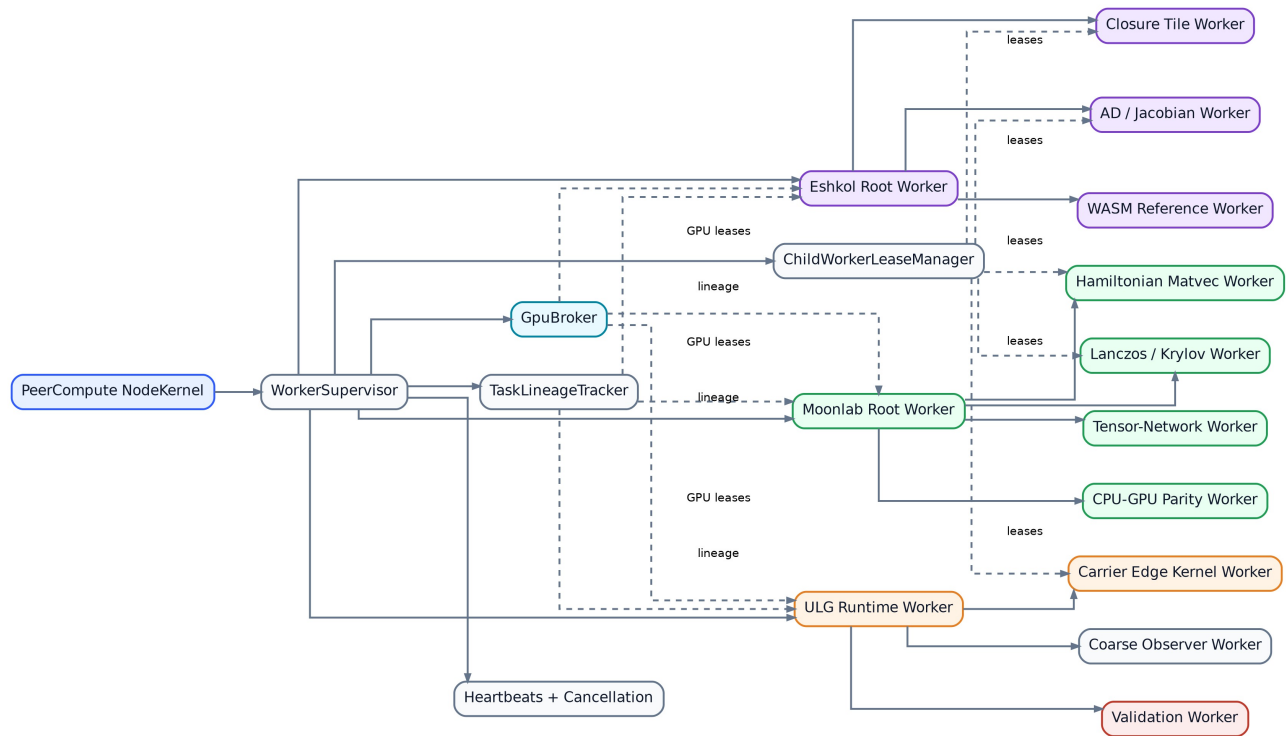


Figure 30: Supervised worker tree

File: 02_supervised_worker_tree.mmd

6.16 03 Closure Factory Flow

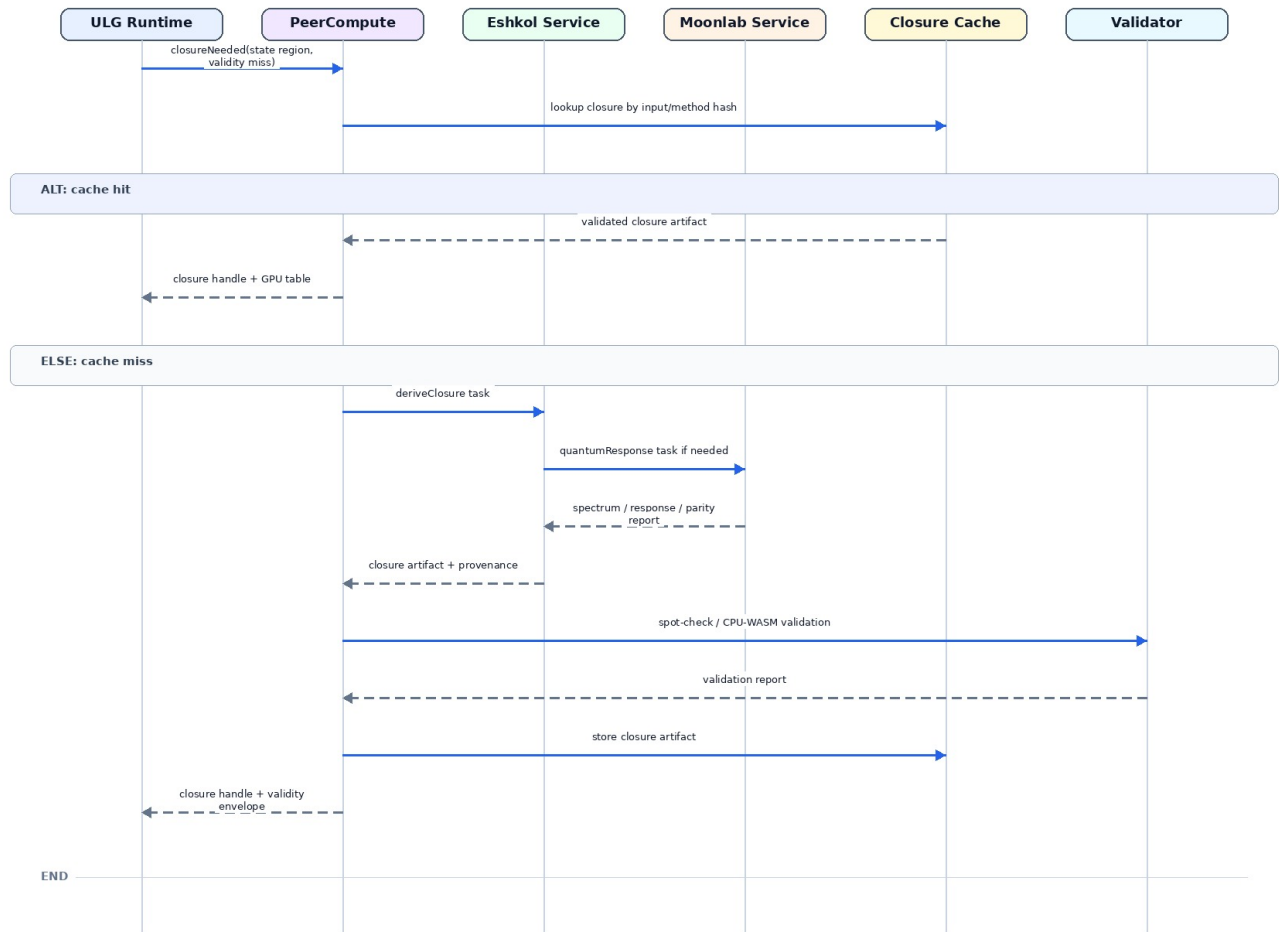


Figure 31: Closure derivation sequence

File: 03_closure_factory_flow.mmd

6.17 04 Shared Gpu Abi

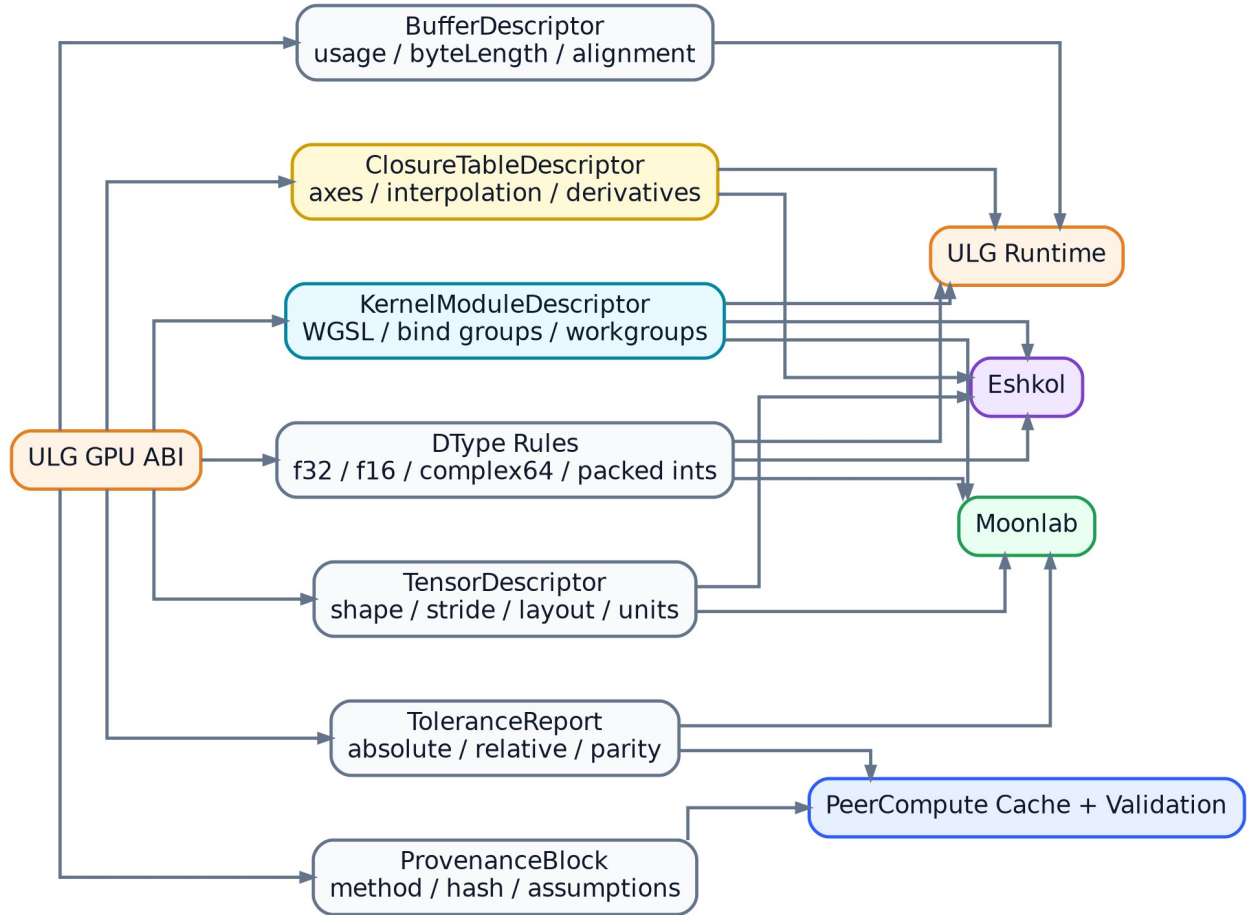


Figure 32: Shared ULG GPU ABI

File: [04_shared_gpu_abi.mmd](#)

6.18 05 Data Lifecycle

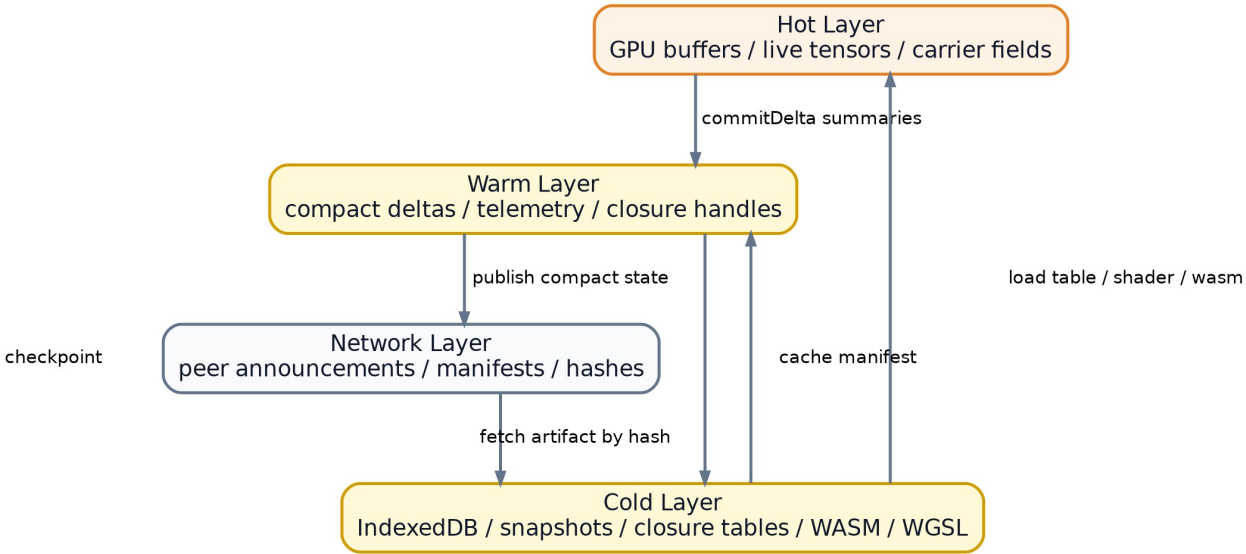


Figure 33: Data lifecycle across hot, warm, cold, and network layers

File: 05_data_lifecycle.mmd

6.19 06 Task State Machine

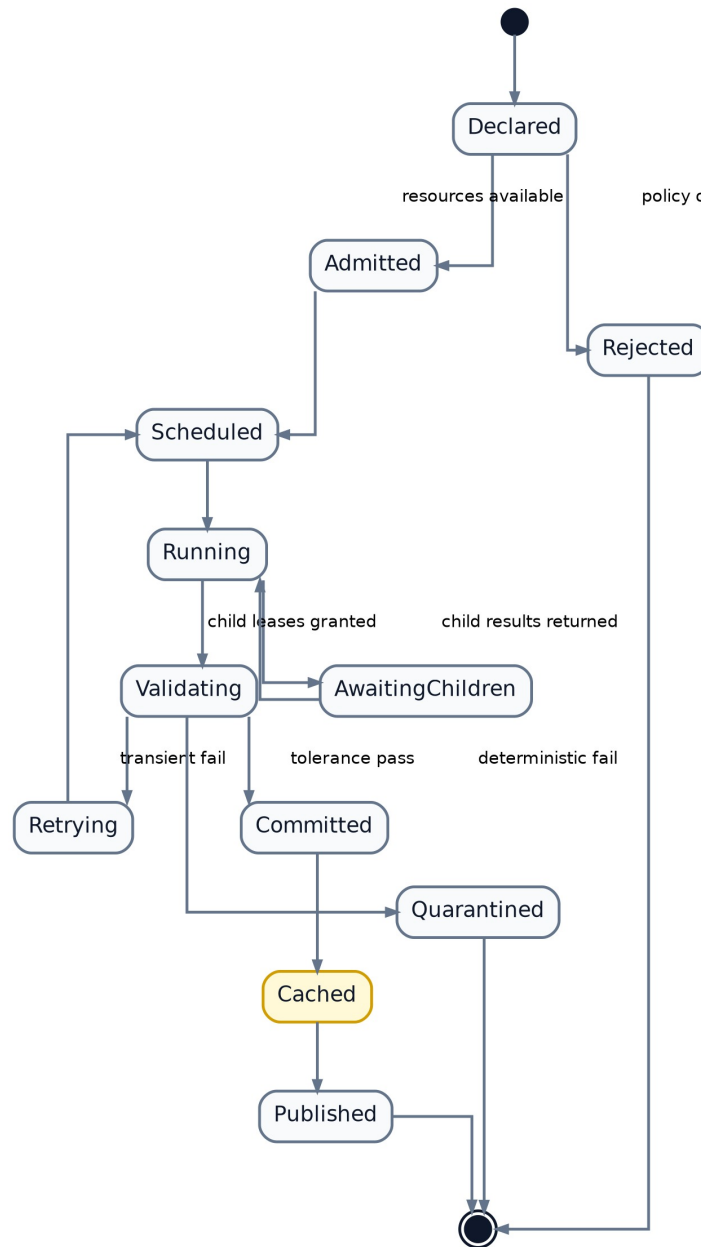


Figure 34: Task lifecycle state machine

File: 06_task_state_machine.mmd

6.20 07 Validation Provenance Loop

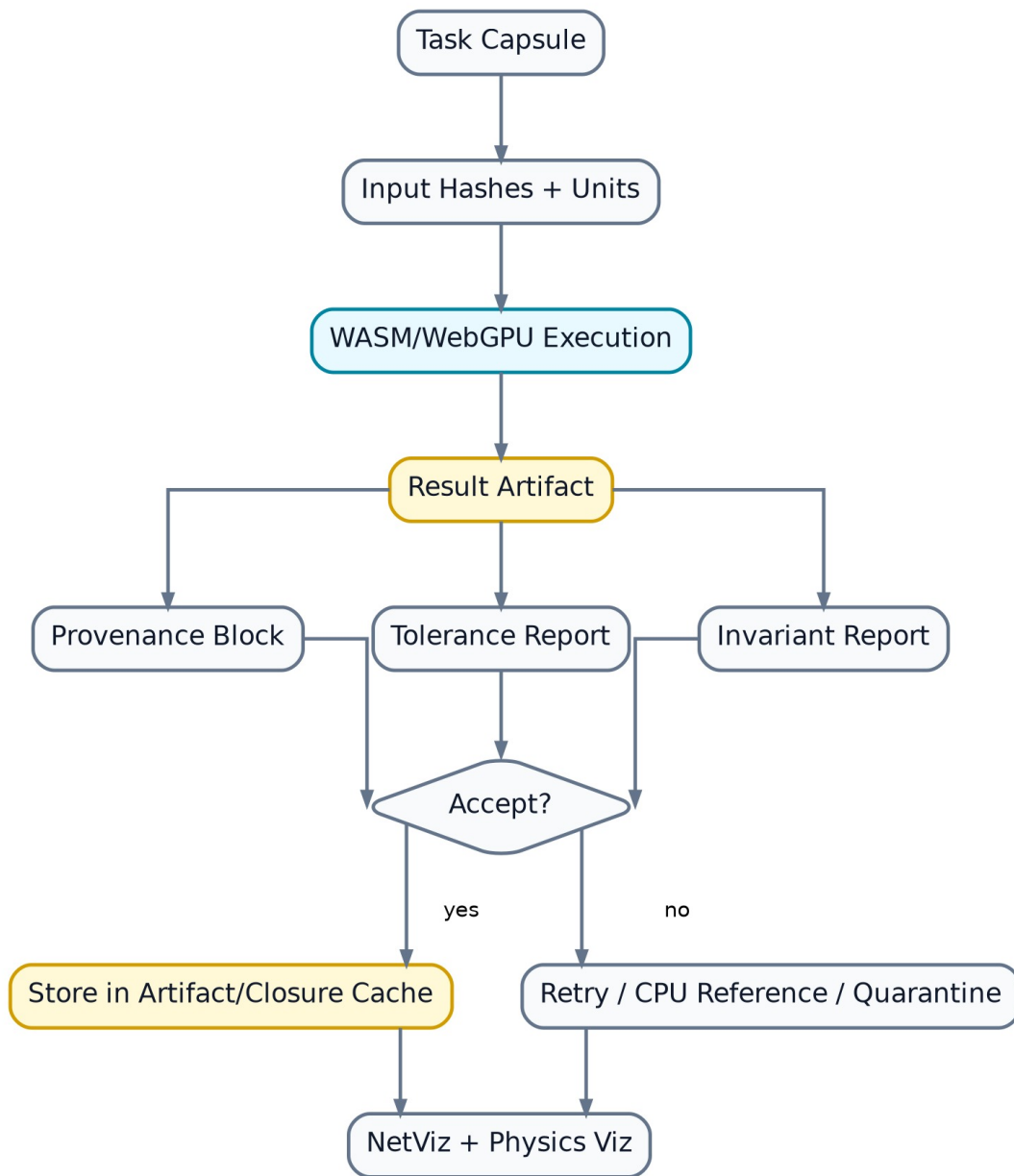


Figure 35: Validation and provenance loop

File: [07_validation_provenance_loop.mmd](#)

6.21 08 Library Extension Map

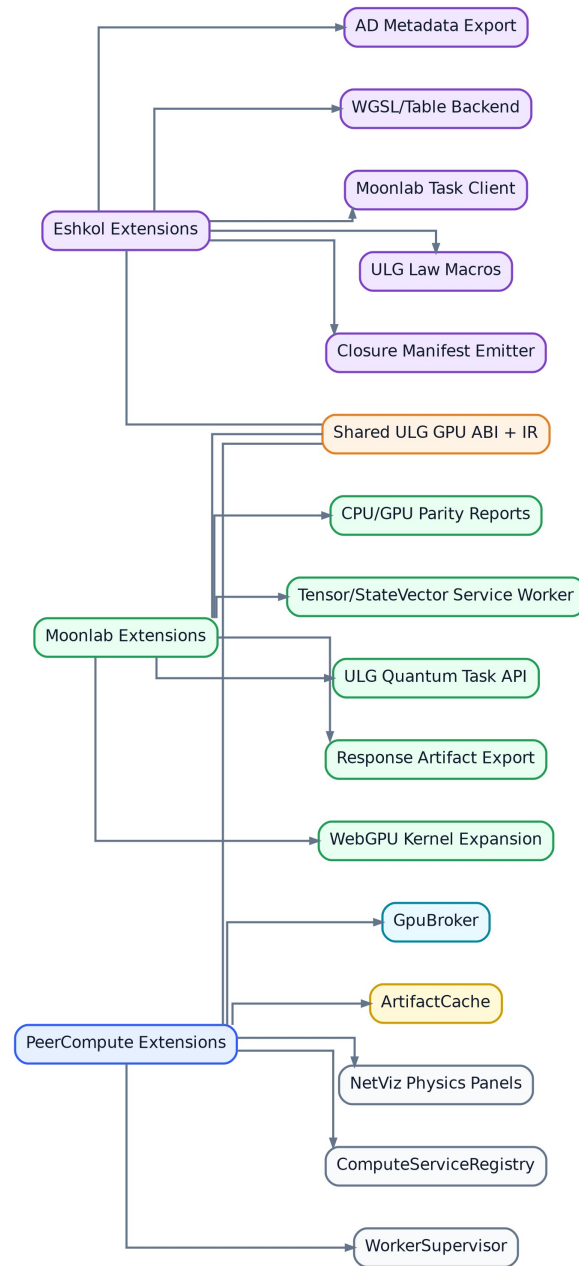


Figure 36: Library extension map

File: 08_library_extension_map.mmd

6.22 09 Roadmap

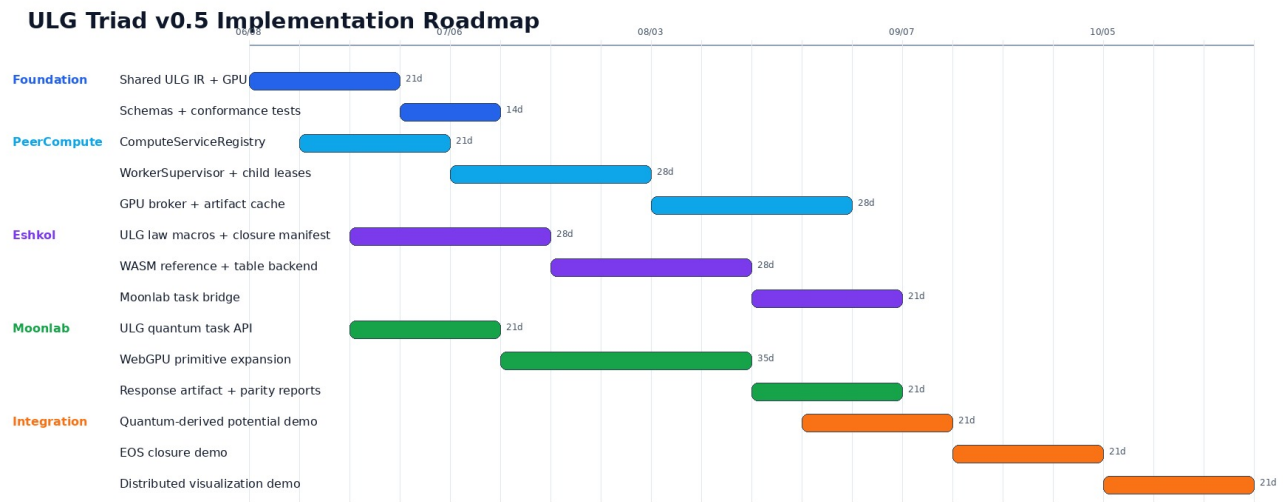


Figure 37: ULG Triad v0.5 Implementation Roadmap

File: [09_roadmap.mmd](#)

6.23 10 Visualization Telemetry

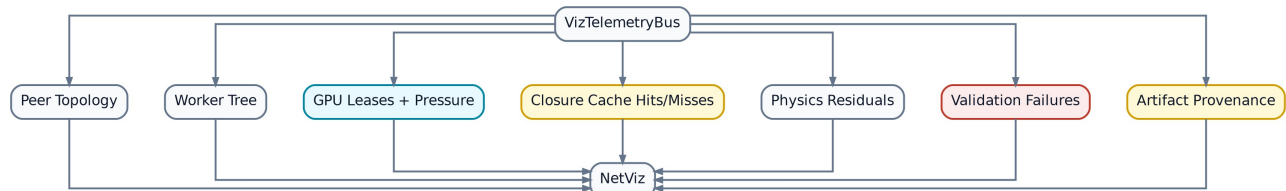


Figure 38: Visualization telemetry bus

File: [10_visualization_telemetry.mmd](#)

6.24 11 Security Resource Model

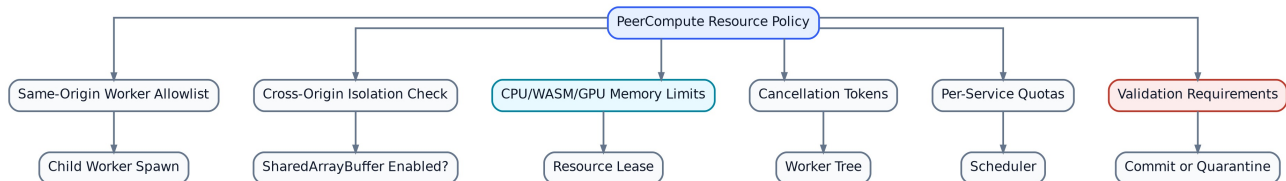


Figure 39: Resource and security policy

File: [11_security_resource_model.mmd](#)

6.25 12 End To End Demo

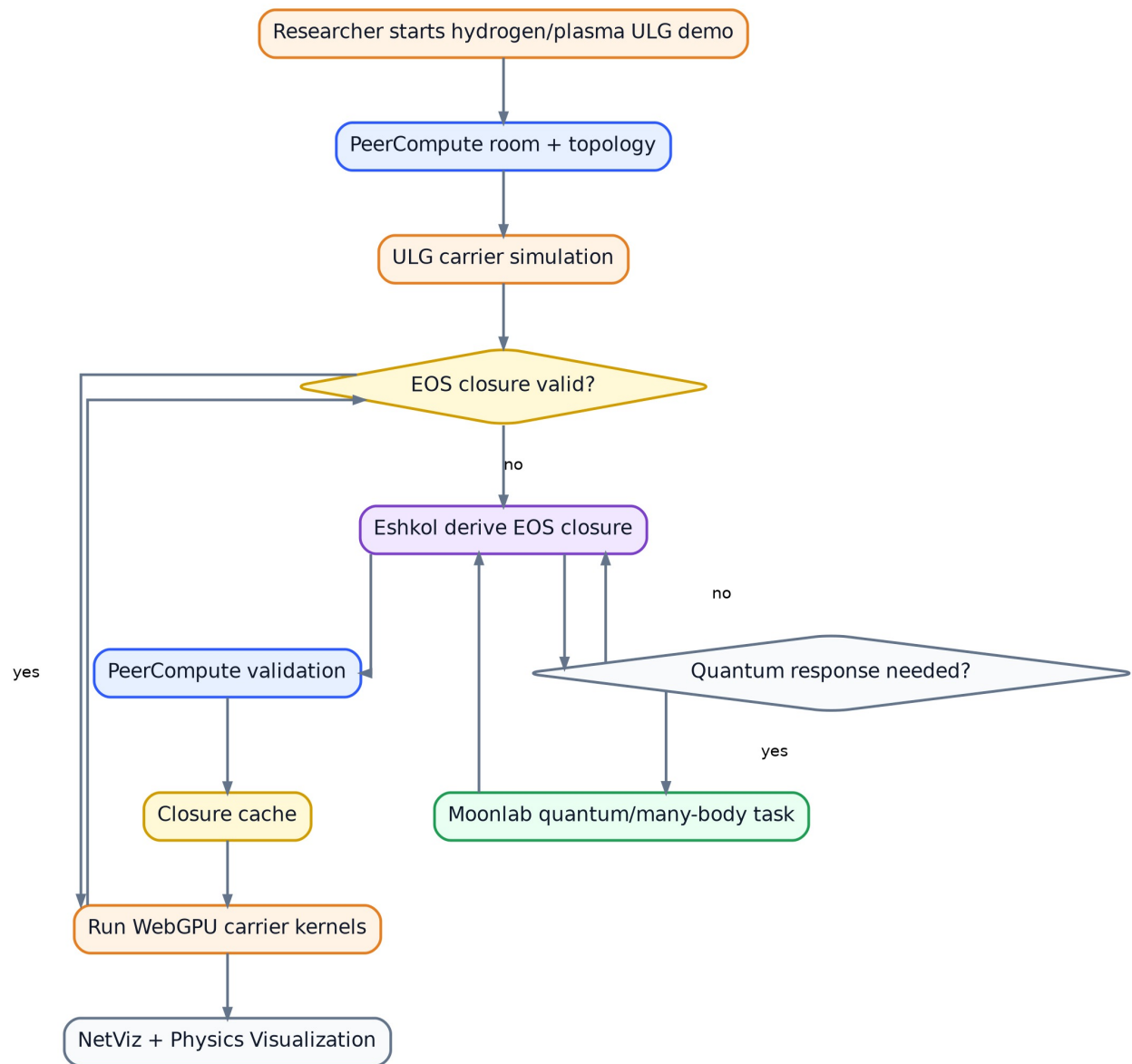


Figure 40: Hydrogen plasma demo flow

File: [12_end_to_end_demo.mmd](#)

7 Appendix B – JSON Schema Sketches

These schema sketches define the initial interchange contracts for compute services, task capsules, closure artifacts, quantum-response artifacts, and validation reports.

7.1 Schema index notes

These schemas are intentionally initial sketches. They exist to make the v0.5 contracts concrete enough to test across PeerCompute, Eshkol, and Moonlab.

- [closure_artifact.schema.json](#)
- [compute_service_manifest.schema.json](#)
- [quantum_response_artifact.schema.json](#)
- [task_capsule.schema.json](#)
- [validation_report.schema.json](#)

7.2 closure_artifact.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ULG Closure Artifact v0.5",
  "type": "object",
  "required": [
    "closureId",
    "sourceService",
    "closureKind",
    "inputs",
    "outputs",
    "execution",
    "validity",
    "provenance"
  ],
  "properties": {
    "closureId": {
      "type": "string"
    },
    "sourceService": {
      "type": "string"
    },
    "closureKind": {
      "type": "string"
    },
    "inputs": {
      "type": "array"
    },
    "outputs": {
      "type": "array"
    },
    "derivatives": {
      "type": "array"
    },
    "execution": {
      "type": "object"
    },
    "validity": {
      "type": "object"
    },
    "uncertainty": {
      "type": "object"
    },
    "validation": {
      "type": "object"
    }
  },
}
```

```

    "provenance": {
      "type": "object"
    }
  }
}

```

7.3 compute_service_manifest.schema.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ULG Compute Service Manifest v0.5",
  "type": "object",
  "required": [
    "serviceId",
    "version",
    "runtime",
    "entry",
    "capabilities",
    "abi"
  ],
  "properties": {
    "serviceId": {
      "type": "string"
    },
    "version": {
      "type": "string"
    },
    "runtime": {
      "enum": [
        "wasm",
        "webgpu",
        "wasm-webgpu",
        "js",
        "native-bridge"
      ]
    },
    "entry": {
      "type": "object",
      "required": [
        "workerModule"
      ],
      "properties": {
        "workerModule": {
          "type": "string"
        },
        "wasmModule": {
          "type": "string"
        },
        "initExport": {
          "type": "string"
        }
      }
    },
    "childWorkers": {
      "type": "object"
    }
  }
}

```

```

    "resources": {
      "type": "object"
    },
    "capabilities": {
      "type": "array",
      "items": {
        "type": "string"
      }
    },
    "abi": {
      "type": "object"
    },
    "validation": {
      "type": "object"
    }
  }
}

```

7.4 quantum_response_artifact.schema.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ULG Quantum Response Artifact v0.5",
  "type": "object",
  "required": [
    "artifactId",
    "sourceService",
    "taskKind",
    "inputHash",
    "method",
    "representation",
    "outputs",
    "validation",
    "provenance"
  ],
  "properties": {
    "artifactId": {
      "type": "string"
    },
    "sourceService": {
      "const": "moonlab"
    },
    "taskKind": {
      "type": "string"
    },
    "inputHash": {
      "type": "string"
    },
    "method": {
      "type": "string"
    },
    "representation": {
      "type": "string"
    },
    "outputs": {
      "type": "object"
    }
  }
}

```

```

    },
    "uncertainty": {
      "type": "object"
    },
    "validation": {
      "type": "object"
    },
    "provenance": {
      "type": "object"
    }
  }
}

```

7.5 task_capsule.schema.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ULG Task Capsule v0.5",
  "type": "object",
  "required": [
    "taskId",
    "rootTaskId",
    "serviceId",
    "taskKind",
    "inputHash",
    "methodHash"
  ],
  "properties": {
    "taskId": {
      "type": "string"
    },
    "rootTaskId": {
      "type": "string"
    },
    "serviceId": {
      "type": "string"
    },
    "taskKind": {
      "type": "string"
    },
    "inputs": {
      "type": "array"
    },
    "outputs": {
      "type": "array"
    },
    "inputHash": {
      "type": "string"
    },
    "methodHash": {
      "type": "string"
    },
    "resources": {
      "type": "object"
    },
    "validation": {

```



```

    "type": "object"
  },
  "provenance": {
    "type": "object"
  }
}
}

```

7.6 validation_report.schema.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ULG Validation Report v0.5",
  "type": "object",
  "required": [
    "status",
    "validationMode",
    "toleranceProfile"
  ],
  "properties": {
    "status": {
      "enum": [
        "unchecked",
        "pass",
        "warn",
        "fail",
        "quarantined"
      ]
    },
    "validatorPeer": {
      "type": "string"
    },
    "validationMode": {
      "type": "string"
    },
    "toleranceProfile": {
      "type": "string"
    },
    "maxAbsError": {
      "type": "number"
    },
    "maxRelError": {
      "type": "number"
    },
    "invariantDrift": {
      "type": "object"
    },
    "notes": {
      "type": "array",
      "items": {
        "type": "string"
      }
    }
  }
}

```

ULG Triad v0.5 Current Status Addendum

Worker-first SPH cold-start, cache, reaction, and gas-pressure slice

2026-06-11

Purpose

This addendum updates the checked-in `ULG_Triad_v0.5_Pretty_Diagrams.pdf` with the current implementation status as of 2026-06-11. The original PDF source was not present in the repository, so the original 68-page PDF is preserved and this status packet is appended as reproducible markdown-generated pages.

Current Status

The ULG browser demo remains an evidence-oriented integration scaffold for PeerCompute, Eshkol, MoonLab, and the ULG runtime. It does not claim calibrated magnetar science, full SPH validation, full phase-change validation, or complete first-principles material solver residency.

Completed since the original v0.5 PDF:

- Strict material-property provenance gates are active in the SPH path.
- Element and compound material closures are derived through the generic lower-level material pipeline instead of reference constants in the default runtime.
- Relativistic/interband optical response is implemented on the CPU and packed into WebGPU-consumable optical rows for the renderer.
- SPH/MLS-MPM state, mechanics, P2G, grid update, G2P, resident stepping, thermal response, reaction conversion, and render-field slices now have explicit ABI records and optional WebGPU paths, with validation flags kept false unless the local parity/evidence actually exists.
- The SPH phase renderer now renders continuous material surfaces, keeps transparent H2O visible, and preserves material/phase render ordering.
- General reaction candidate discovery now handles balanced element/compound pairs such as Li/H2O, Na/H2O, Cs/H2O, Na/Cl, Na/Cl2, Mg/O2, Fe/O2, and Fe/H2O without the old Na-only branch.
- The live SPH overlay shows independent render, physics, and resident FPS counters plus WebGPU/CPU fallback warnings.

Cold-Start Cache Slice

The first cold-start cache/performance slice is complete.

Implemented:

- `discoverReactions()` no longer disables its cache when `materialProperties` is provided. It now includes stable material-property provenance hashes in the cache key.
- Reaction discovery can return `memory-cache-hit`, `persistent-cache-hit`, or `derived-cache-miss` status.
- Persisted reaction records and product reuse records are written to an SPH cold-start localStorage cache namespace guarded by generator fingerprints.
- Product closures supplied by the cache are reused during reaction discovery before deriving a new product closure.
- Initial SPH material/reaction/view-state rebuilds run through the supervised `ulg-runtime` worker when available, so cold first-principles derivation does not block initial UI construction.

- The SPH overlay exposes `cold cache`, `cache clear`, and `perf trace` status rows.
- A scoped `Clear Cache` button clears ULG SPH derived material and cold-start cache families plus the in-memory reaction discovery cache.
- When render FPS drops below 30 during active CPU closure work, the warning banner reports `deriving material or reaction properties`.
- The ultra-low-FPS cache-miss auto-pause path was removed; the long 0.1 FPS cold-start period remains a performance target to fix by reducing or offloading work.
- Manual Step/Play can reconstruct the interactive driver from cache after the worker prepopulates material/reaction closures, preserving the existing reaction-step path without forcing cold derivation during initial load.
- The follow-up cold-start slice consumes partial material cache hits instead of discarding all cached closures when one runtime default is missing.
- Missing required runtime materials are derived individually through the generic derivation path instead of forcing a full default closure rebuild.
- `ulg-runtime` is pre-spawned with the other demo services.
- Worker rebuild, resident MLS-MPM, and scene particle-sync stages now expose timing breakdowns in the overlay.
- Deterministic thermal material tables, thermal closure graph banks, phase response tables, optical/PBR rows, reaction tables, and GPU warmup signatures are written into the cold-start cache as derived artifact records.
- Balanced stoichiometric discovery now preserves reactant/product term arrays and rejects stale persistent reaction records that lack them.
- The CPU reference reaction path can produce multiple products. $\text{Na} + \text{liquid H}_2\text{O}$ now produces NaOH plus H_2 in the browser demo.
- The SPH demo now reports sealed-box gas pressure diagnostics, including H_2 partial pressure from reaction byproducts.

Remaining:

- Rehydrate persisted static table rows into scene/WebGPU upload paths on warm reload.
- Move large cache serialization/parsing and remaining resolver work farther out of the UI thread and into workers, PeerCompute state, or WebGPU kernels.
- Add stale-record browser probes and measured cold/warm/clear/repopulate startup deltas.
- Mirror cacheable derived closures into PeerCompute state so peers can reuse valid first-principles derivation artifacts.

Current Validation

Validated locally on 2026-06-11:

- `node --test tests/reactionDiscovery.test.mjs`: 8/8 pass.
- `node --test tests/reactiveChemistry.test.mjs`: 7/7 pass.
- `node --test tests/sphPhaseDemo.test.mjs`: 7/7 pass.
- Focused contract, renderer, and chemistry tests: 16/16 pass.
- Focused HTTPS Chromium SPH browser tests against `https://127.0.0.1:5173`: 3/3 pass.
- Focused HTTPS Chromium $\text{Na}/\text{H}_2\text{O}$ pressure test: 1/1 pass.
- `npm test`: 336/336 pass.
- `npm run build`: pass with the existing Vite large-chunk warning.
- `npm run build:pages`: pass with the existing Vite large-chunk warning; `docs/` regenerated.

Active Risks

- The broad material/optics/reaction derivation stack is still CPU-heavy. The current worker-first path protects initial UI responsiveness but does not make every resolver WebGPU-resident yet.
- Reaction energetics still include provisional candidate cases when crude generated geometries give bad signs. CPU reference multi-product conversion and gas-pressure diagnostics exist, but full reaction extent solving, validated free-energy refinement, and WebGPU resident multi-product execution remain active todo items.

- Steam optics are not yet phase/state keyed enough to distinguish vapor from condensed droplets in a general optical closure cache.
- Nuclear, radiation, Cherenkov, fission, fusion, and radioactive decay closures remain planned but not implemented.

Immediate Next Work

1. Rehydrate static table cache rows into warm scene/WebGPU uploads and move cache serialization/parsing off the UI thread.
2. Promote balanced multi-product reaction execution into a full extent solver and then into the resident WebGPU reaction kernel.
3. Add phase-resolved H₂O/steam optics keyed by phase, temperature, pressure, density, and droplet fraction.
4. Continue migrating material property resolvers to worker/WebGPU-resident execution while keeping CPU reference parity and provenance checks.